BIOL 336 REVIEW SHEET

Note: This sheet does not include every lecture

Part I. Intro, Darwin's Dangerous Idea and Modern Synthesis

Lecture 1: Introduction to Evolution

1. The World Before Darwin

A. Why Diversity & design apparent in nature:

① Before Evolution:

- Plato:
    - Theory of Forms
    - The physical world is an imperfect reflection of a perfect idealised world
- Aristotle:
    - hierarchical classification, *Scala Naturae*, to divide lifeforms into groups: Man/Mammals with 4 limbs/Birds/Plants/Minerals…
    - Became "Great Chain of Being" in Medieval——Natural theology, biology as a way of studying the mind of the Creator
- Natural Theology: Species mostly immutable
    - **John Ray**: defined species as groups distinguished by traits that remain stable across generations. (the first purely biological definition of species)
    - **Carl Linnaeus**: regularized the binomial system for naming species: a genus name + a specific epithet
    - **William Paley**: the complexity and purposeful design of nature implies the existence of a divine creator

② Pre-Darwin evolutionists:

- Erasmus Darwin

    - All life comes from a common ancestor (“one living filament”).
    - Species change over time as organisms adapt to their environments (transformism)
- Jean-Baptiste Lamarck
    - Species progress from simple forms to increasingly complex forms; new simple organisms arise by spontaneous generation.
    - **First law**: Use or disuse of a structure leads to its development or diminishment
    - **Second law**: These acquired characters can be passed on to offspring
- David Hume vs Patrick Matthew
    - **Hume**: Recognized that natural selection could explain design in nature, but thought it was unlikely to be the main explanation.
    - **Matthew**: Correctly identified natural selection occurs, but did not realize its broad explanatory power.
- Two key influences to Darwin:
- Thomas Malthus
    - Principle of Overproduction: organisms produce more offspring than can survive.
    - Populations grow geometrically, while food supply grows linearly, causing inevitable competition.
    - Poverty, disease, and famine result from this struggle for existence.
- Charles Lyell
    - Uniformitarianism: the Earth is shaped by slow, ongoing processes (e.g., erosion), not sudden catastrophes.
    - Introduced the idea of Deep Time — the Earth is very old.

- Came close to evolutionary thinking, but limited biological change to the "natural wants and propensities" of species.

2. What Darwin Did

A. the Voyage of the Beagle 1831-1836

- Similar large flightless birds (rhea, ostrich, emu) in similar habitats were different on different continents, suggesting species change with geography.
- Neighboring regions often had closely related but distinct species that coexisted and competed.
- On the Galápagos, tortoises and finches varied island to island, showing local adaptation and divergence.

B. The Darwin-Wallace Incident

- In 1858, Wallace independently discovered natural selection and sent his ideas to Darwin.
- **Darwin–Wallace Hypothesis**, the first public announcement of evolution by natural selection, presented at the Linnean Society on July 1, 1858

C. The Origin of Species

Key Arguments in:

- 2 key assumptions: ① individuals vary from one another; ② and offspring resemble their parents
- Variation & heritability lead to: the struggle for survival gives rise to adaptation by natural selection
- Concrete Evidence: domestication

D. The world after Darwin

- Diversity & apparent design in nature — still true
- Clearly not random – there must be some order to it — still true

- The order reflects *evolution* (the tree of life)
- Natural *history* – biology as a way of studying the process of evolution
- Species *change*

Lecture 2: New Modern Synthesis

A. Outline

Three types of objection to evolution by natural selection:

a) Motivated

b) Logical

c) Empirical

The Modern Synthesis:

a) Empirical Discoveries

b) The Birth of Evolutionary Theory

B. Recap: Darwin's argument and its significance

1. There is empirical evidence for two key assumptions: individuals vary from one another; and offspring resemble their parents
2. Variation & heritability lead to an inevitable process: the struggle for survival gives rise to adaptation by natural selection
3. There is evidence for this process in front of our eyes from domestication
4. This process can explain the patterns of natural history

C. Aftermath

① Motivated Objections to Darwin

- Often framed in religious terms
- But perhaps more to do with authority than theology

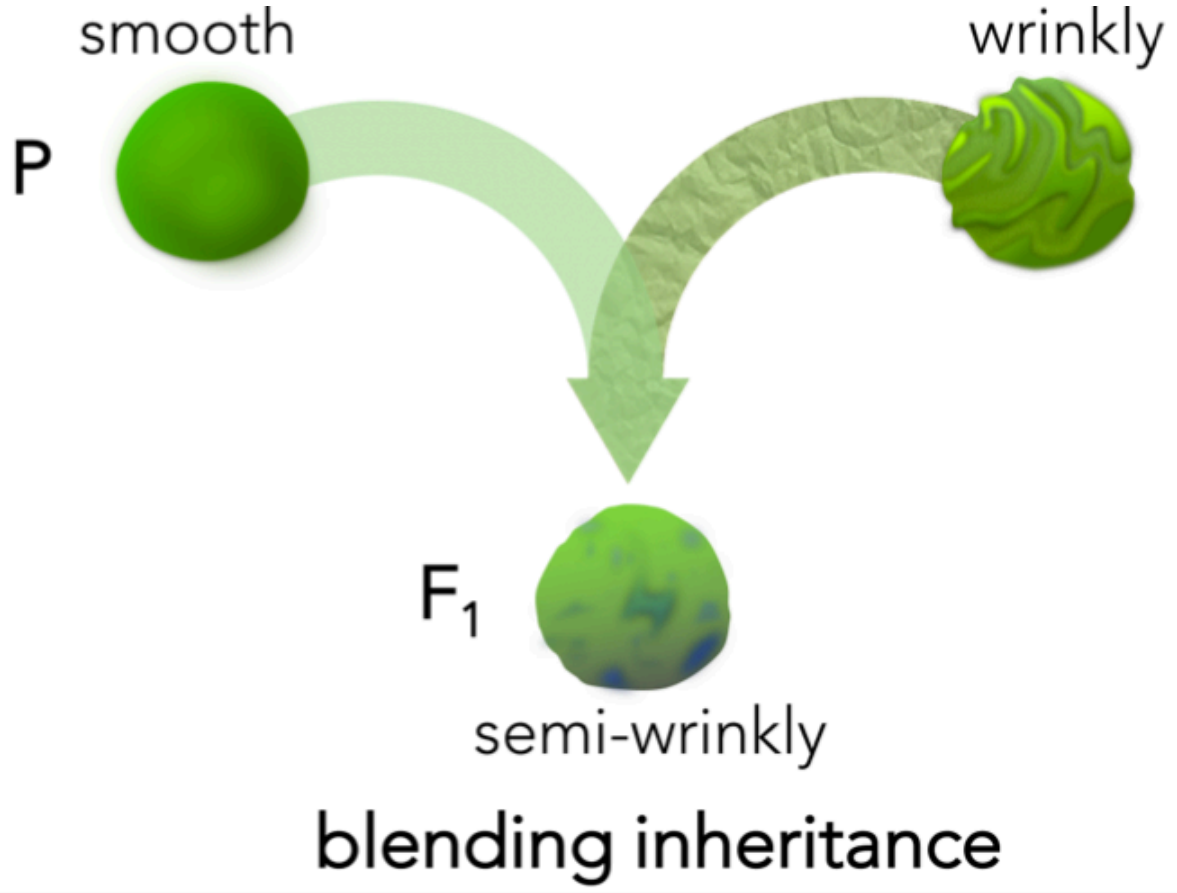
"The Great Debate":

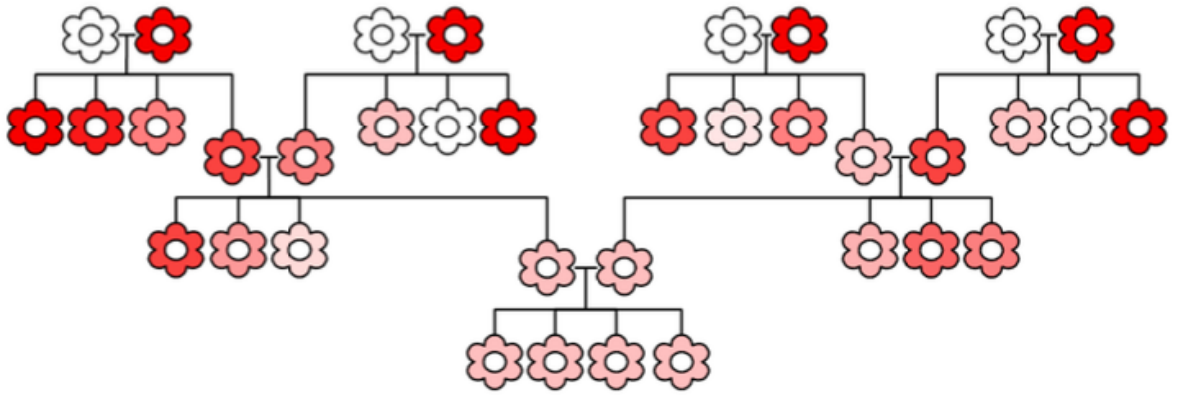
- Often centered around public sentiment (hence theatrics)

- Motivated reasoning (hence logical fallacies) • Evolution seen as a threat to authority?
- Darwin left mostly to others
- Set the context for other types of objection

② Logical Objections to Darwin

- Key example: Jenkin's Cannon
    - [Darwin's argument] seems no more accurate than to conclude that because we observe that a cannon- ball has traversed a mile in a minute, therefore in an hour it will be sixty miles off, and in the course of ages that it will reach the fixed stars."
    - Jenkin's:
    - Evolution by natural selection cannot account for life on Earth unless we know how fast the cannonball goes
    - Empirical argument: Most domesticated species slow down in their rate of change
    - Logical argument: Blending inheritance must slow evolutionary change
- Blending inheritance:





Even if red flowers are highly beneficial, they can never spread from rarity to fixation by natural selection

③ Empirical Objections to Darwin

- Key example: Mendelian Genetics

- Mendelian Genetics as an Alternative
    - Hugo de Vries (1890s–1900s:
    - Rediscovered Mendel’s work.

        - Studied evening primrose; saw sharply different forms.
        - Introduced “mutation” for sudden, discontinuous variation.
        - Proposed mutation theory: big jumps create new forms → evolution.
        - Became the main Mendelian challenge to Darwin’s gradualism.
- Mendelian View (Mendelians)
    - Nature shows discontinuous variation (Mendelian patterns).
    - Supports Jenkin’s argument: small variations are ineffective for selection
    - Therefore real evolutionary change requires large mutations, not slow natural selection.
    - Rejects Darwin’s claim of continuous natural variation.
- Goldschmidt & “Hopeful Monsters”
    - Richard Goldschmidt Inspired by developmental mutants (e.g., Aristapedia: antenna → leg).
    - Proposed “Hopeful Monsters”: single, large-effect mutations produce new species.
    - Claimed species origins require processes different from small within-species changes.

D. Mendelism vs Darwinism

- Mendelians
    - Relied on laboratory experiments, discontinuity.
    - Rejected natural selection as the driver of evolution.
- Biometricians
    - Used natural observations, continuous variation.
    - Supported Darwinian gradualism.
    - Mendel’s results oversimplified, not universal

	Motivated	Logical	Empirical
Illustrative Example	e.g. The Great Debate	e.g. Jenkin's Cannon	e.g. Bateson vs Weldon
Basis	Motivated reasoning	Logical flaws	Evidence
Effect on Darwin	Darwin left to others	Troubled Darwin	Mostly post-Darwin

E. The Modern Synthesis: Empirical Discoveries in 1900s

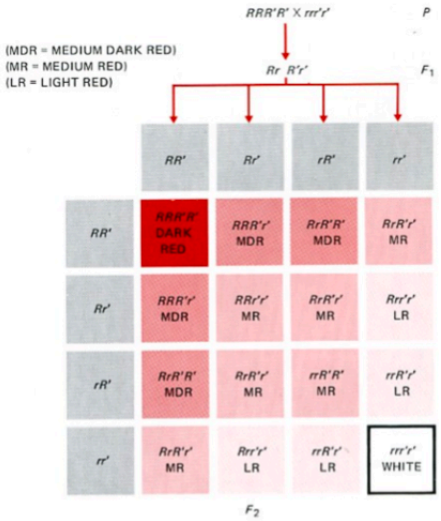
- Inheritance was particulate, not blending
- Mendelian inheritance could be consistent with can be consistent with biometrics

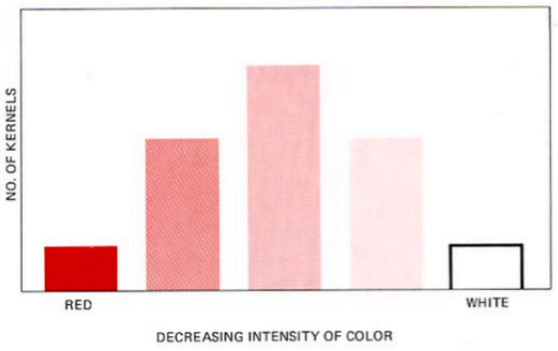
① William Castle (1914)

- Studied a continuous trait: rat coat coloration ("hood").
- Performed selection for more and less pigmentation.
- Demonstrated that continuous selection on existing variation can produce phenotypes beyond the original range.
- → Shows small variations can accumulate progressively (supports Darwinian gradualism).

② Nilsson-Ehle (1908)

- Studied continuous variation in wheat kernel colour.
- Showed that two gene pairs (with no dominance) following Mendelian inheritance can generate a continuous distribution of phenotypes.
- → Demonstrates how polygenic inheritance creates continuous traits.





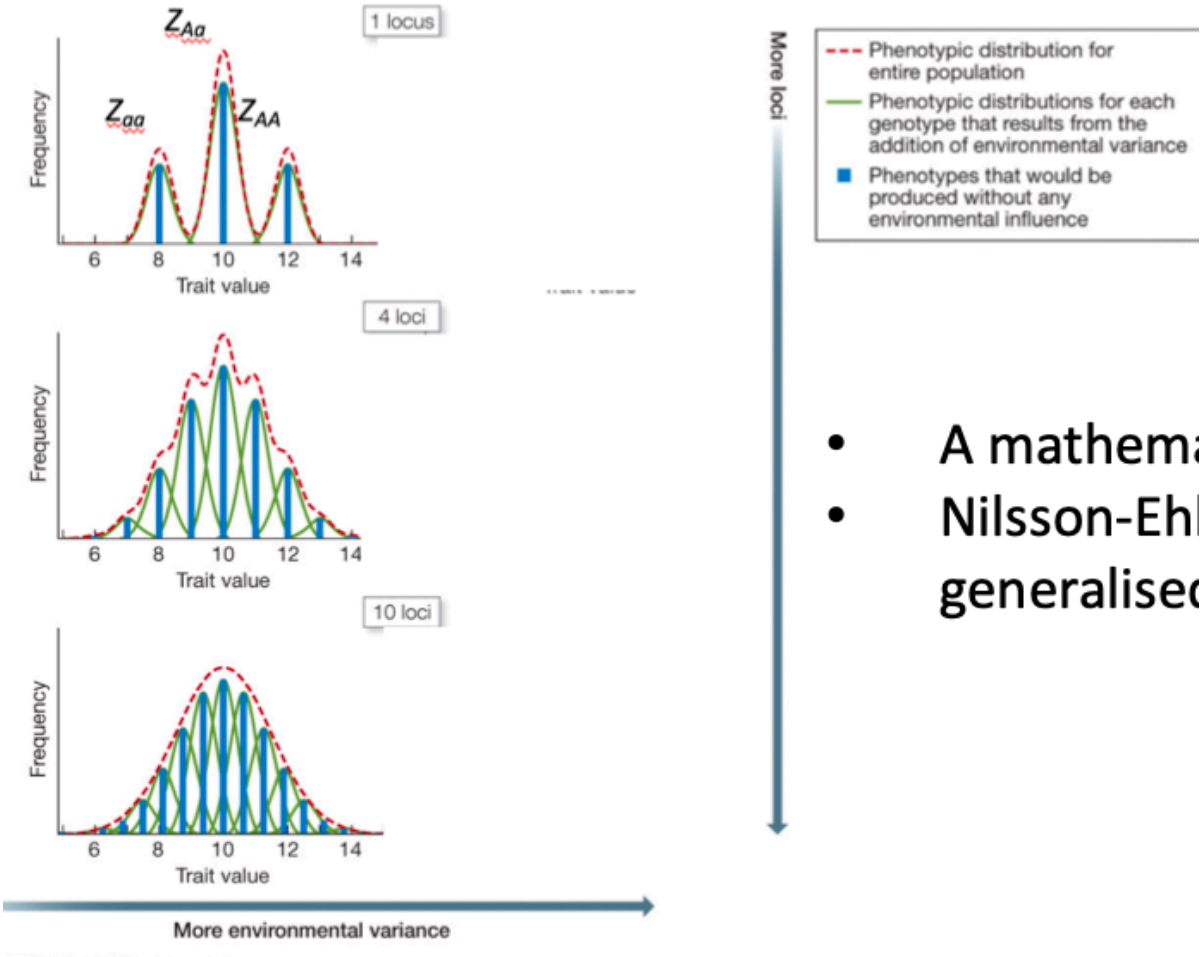
③ Thomas Hunt Morgan (1908)

- Discovered:
    - Sex-linked inheritance
    - Genetic linkage
    - Crossing-over
- Showed that variation in inheritance patterns reflects the physical arrangement of loci on chromosomes.
- → Helped establish the chromosomal theory of inheritance.

F. Origins of population genetics: 1920s-1940s

① Fisher's 1918 Proof of many loci

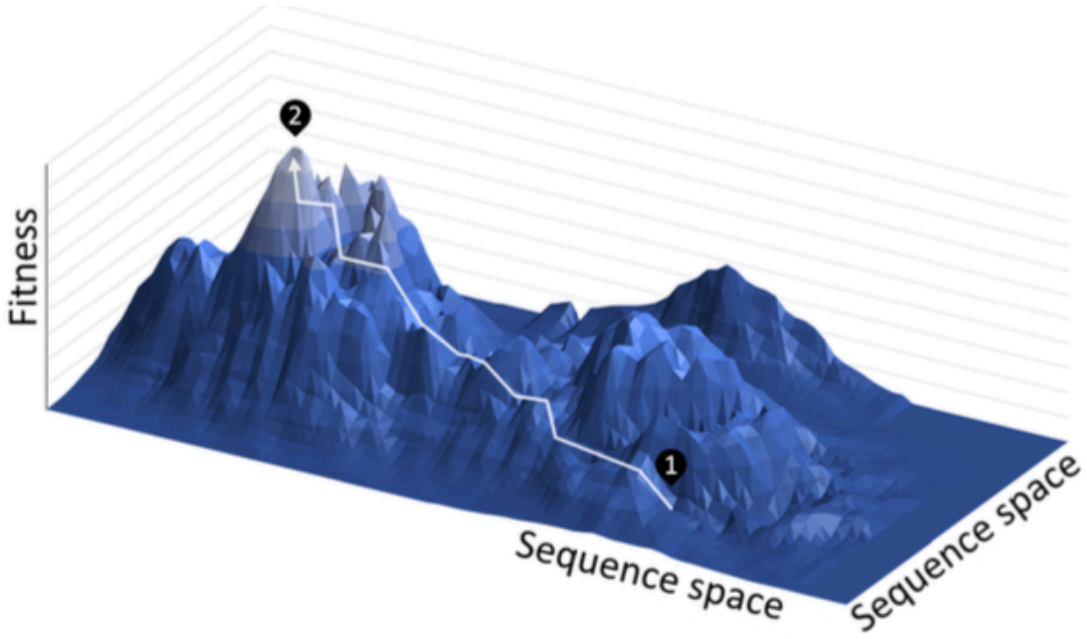
- Nilsson-Ehle's results generalised



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generalised

② Wright: Fitness Landscapes

- 3-D metaphor:
    - x–y: genotypes
    - z: fitness
- Peaks = high fitness; valleys = low fitness.
- Populations “climb” peaks by natural selection.
- Drift can push populations off peaks → across valleys → to new peaks.



③ JBS Haldane: Applications

a) Linked theory to data (e.g. 1908-1912 first demonstration of genetic linkage in mammals)

b) Fundamental results in population genetics (e.g. probability of fixation)

c) Linked population genetics to human issues (e.g. sickle cell anaemia)

Part II. POPULATIONS and POPULATION GENETICS

1. Populations 1 & 2: Introduction to population genetics (haploid selection)

A. Evolution by Natural Selection

① Occurs whenever:

- Individuals vary in some trait (VARIANCE)
- Individuals with some trait values are more likely to live and/or reproduce (SELECTION)
- Parents have offspring with similar trait values (HERITABILITY)

- E.g., evolution of COVID-19
    - VARIANCE – introduced by mutations

- SELECTION – “Alpha” (B.1.1.7) variant transmitted at a 40-80% higher rate than previous viruses
- HERITABILITY – provided by RNA replication of this virus

B. Selection in Short Terms:

- An haploid has 2 alleles, A and a
  - Frequencies of A & a are p[t] & q [t] at time *t*.
  - $p[t] + q[t] = 1$
- Individual haploids carrying these alleles have different fitness:
  - W_A =fitness of A
  - W_a = fitness of a
  - Fitness = The average contribution per parent to the next generation, including survival and reproduction.
- Combined:
  - Frequency of A before selection: $p[t]$
  - Frequency of A after selection (fitness applied): $\frac{W_A\, p[t]}{W_A p[t] + W_a q[t]}$

$$p[t+1] = \frac{W_A\, p[t]}{W_A p[t] + W_a q[t]}$$

- This leads to evolutionary change across one generation:

Change ("Δ") in p from generation t to t+1.

$$\Delta p[t] = p[t+1] - p[t] = \frac{(W_A - W_a)p[t]q[t]}{W_A p[t] + W_a q[t]}$$

- $p[t]\ q[t]$ represents the haploid genetic variance.

- Identical equations for evolutionary change for sexual or asexual haploids, as long as there is no selection in the diploid phase.
- Evolution by natural selection depends *only* on the relative fitness of the alleles, not their absolute values.
  - Example: p[t]=⅔, $W_A = 3$, $W_a = 2$, p[t+1]=¾;
    - If very individual has twice as many offspring, so $W_A = 6$ and $W_a = 4$? p[t+1] still be ¾

C. Selection in Long Terms:

① Starting at generation t=0

- G1: $\frac{W_A\, p[0]}{W_A p[0] + W_a q[0]}$
- G2: $p[1] = \frac{W_A\, p[0]}{W_A p[0] + W_a q[0]}$
- G3: $p[2] = \frac{W_A\, p[1]}{W_A p[1] + W_a q[1]}$

- Plug in p[1]

$$p[2] = \frac{W_A\left(\frac{W_A\,p[0]}{W_Ap[0] + W_aq[0]}\right)}{W_A\left(\frac{W_A\,p[0]}{W_Ap[0] + W_aq[0]}\right) + W_a\left(1 - \frac{W_A\,p[0]}{W_Ap[0] + W_aq[0]}\right)}$$

$$p[2] = \frac{W_A^2\,p[0]}{W_A^2\,p[0] + W_a^2\,q[0]}$$

and tidy up:

$$p[3] = \frac{W_A^3\,p[0]}{W_A^3\,p[0] + W_a^3\,q[0]}$$

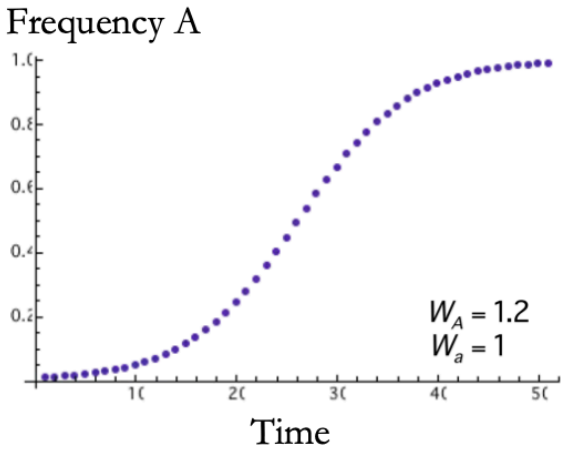
- G4:

- And we got the equation:

$$p[t] = \frac{W_A^t\,p[0]}{W_A^t\,p[0] + W_a^t\,q[0]}$$ | after t generations

②Allele fixation:

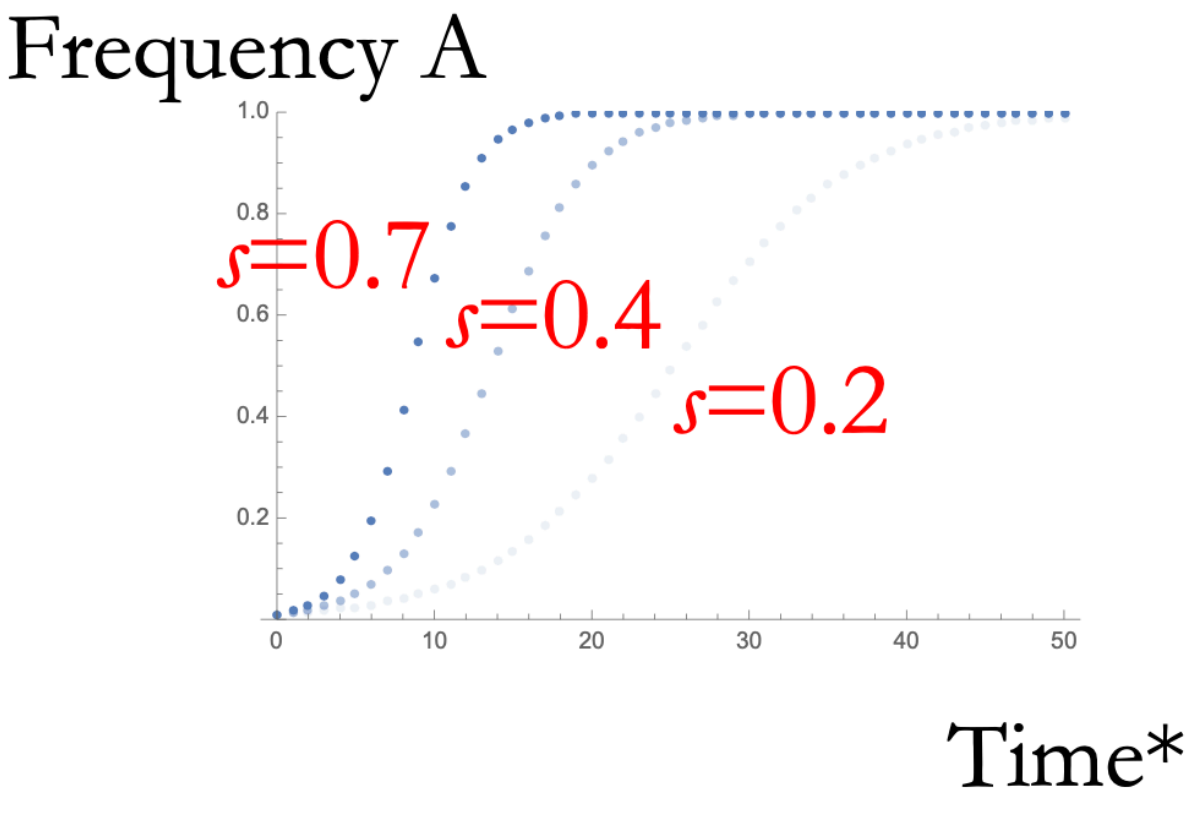
- If $W_A = W_a$, then $p\,[t]$ remains at $p\,[0]$ “Neutral”
- If $W_A \neq W_a$, Most fit allele fixes
  - If $W_A > W_a$, then $p[t] \rightarrow 1$ “Directional selection” favoring A
  - If $W_A < W_a$, then $p[t] \rightarrow 0$ “Directional selection” favoring a

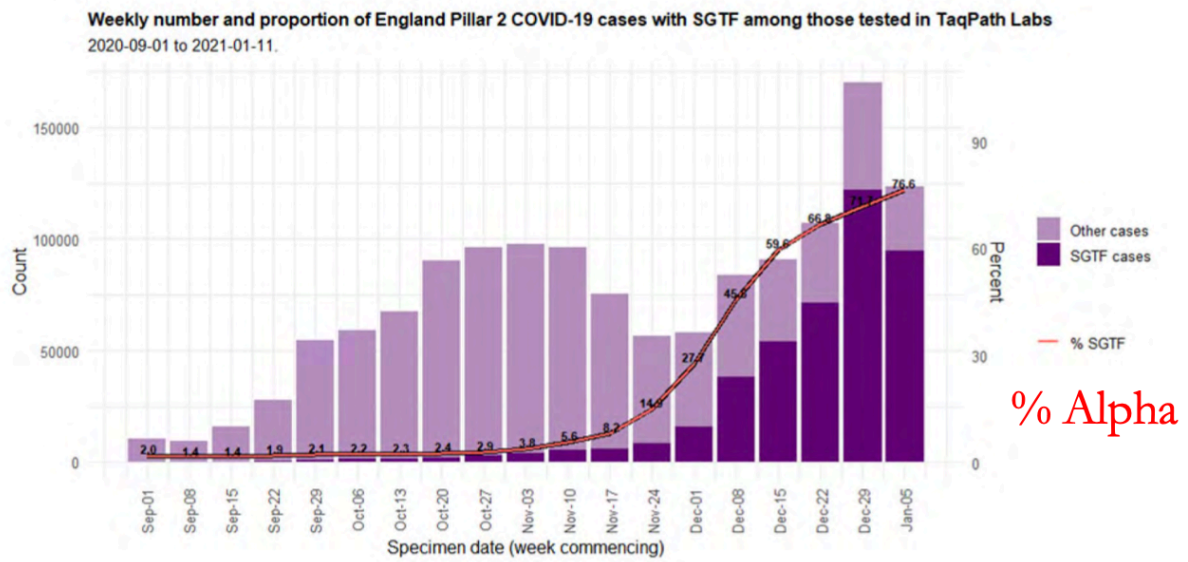


Example: Plot of change over time in the frequency of an allele *A* that increases fitness by 20%.

$$p[t] = \frac{W_A^t\, p[0]}{W_A^t\, p[0] + W_a^t\, q[0]}$$

- When considering the spread of a new beneficial allele (A) in a population of wildtype alleles (a), we typically measure fitness relative to the wildtype:
  - $W_A = 1$
  - $W_a = 1 + \boldsymbol{s}$
    - s is known as the "selection coefficient," which is the proportional increase in fitness caused by replacing allele a with A.
  - Example:
    - the estimated selective advantage of the Alpha variant of COVID is s = 0.4-0.7 (a 40% to 70% increase in fitness)





- ROUGHLY, for allele A to rise from low frequency to high frequency, it takes about:
    - 100 generations if s is around 0.1
    - 1000 generations if s is around 0.01
    - 10,000 generations if s is around 0.001

- As a rule of thumb, if s is 10 times smaller, it takes ~10 times longer to observe the same amount of frequency change.

D. Mean (=average) Fitness

$$\Delta p[t] = \frac{(W_A - W_a)p[t]q[t]}{W_A p[t] + W_a q[t]}$$

Mean fitness, $\bar{W}[t]$

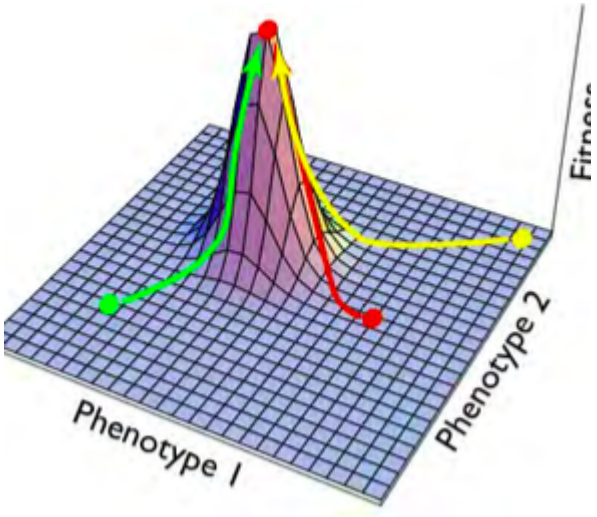
- Mean fitness change across one generation:

$$\Delta\bar{W} = \bar{W}[t+1] - \bar{W}[t] \quad \star$$

$$= \frac{(W_A - W_a)^2 p[t]q[t]}{W_A p[t] + W_a q[t]} \geq 0$$

E. Evolution NOT by natural selection

① If natural selection so effective, Why isn't everything the same?



- Reasons:
    - Organisms vary
    - Environments differ
    - The world is changing

② Evolution NOT by natural selection

- Mutations arise
- Chance plays a role
- Sex and recombination alter
- Alleles favored for effects on some traits may affect other traits (pleiotropy)
- Neighboring alleles in the genome can be dragged with selected alleles (hitchhiking)

Key points

- If $W_A = W_a$, allele A will stay constant in frequency (neutral).
- If $W_A > W_a$, allele A will rise in frequency to fixation (p = 1).
- If $W_A < W_a$, allele A will decline in frequency to loss (p = 0).
- Spread of a beneficial allele always follows an S-shaped curve, increasing slowly when the allele is rare and also when it is common.
- Weakly favored alleles take longer to spread than strongly favored alleles.

- When selection (s) is 10 times weaker, it takes about 10 times longer to observe the same amount of frequency change.

2. Populations 2 and 3: Modelling selection in diploid populations

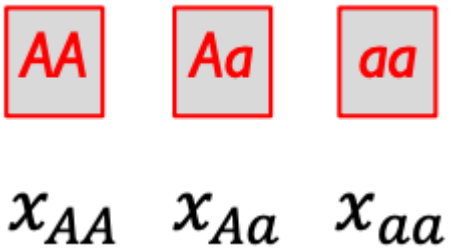
A. Diploid Selection

- Diploid: Two copies of each gene
    - The haploid parents "breed true" –> A-bearing haploid parents give rise to A-bearing offspring, whether they reproduce sexually or asexually.
    - The diploids *only* reproduce asexually breed true (e.g., Aa parents producing Aa offspring).
        - With sexual reproduction, two Aa parents produce a variety of offspring (AA, Aa, and aa).

B. Asexual Production of Diploids

Natural selection acts during the diploid stage:

- W_{AA} = fitness of AA
- W_{Aa} = fitness of Aa
- W_{aa} = fitness of aa



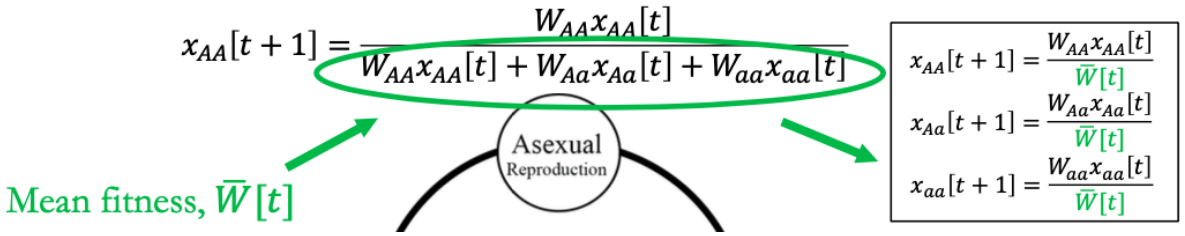
After selection:



- Then calculate fraction of each genotype:

$$x_{AA}[t+1] = \frac{W_{AA}x_{AA}[t]}{W_{AA}x_{AA}[t] + W_{Aa}x_{Aa}[t] + W_{aa}x_{aa}[t]}$$

- Example: "Heterozygous advantage", where heterozygotes have five times the relative fitness, $W_{AA} = 0.2$, $W_{Aa} = 1$, $W_{aa} = 0.2$, $x_{AA}[t] = 0.16$ $x_{Aa}[t] = 0.48$ $x_{aa}[t] = 0.36$
    - $x_{AA}[t+1] = 0.055$ $x_{Aa}[t+1] = 0.822$ $x_{aa}[t+1] = 0.123$
- In asexual populations, mean fitness increases over time (or stays the same)



C. Diploid Selection in Sexuals

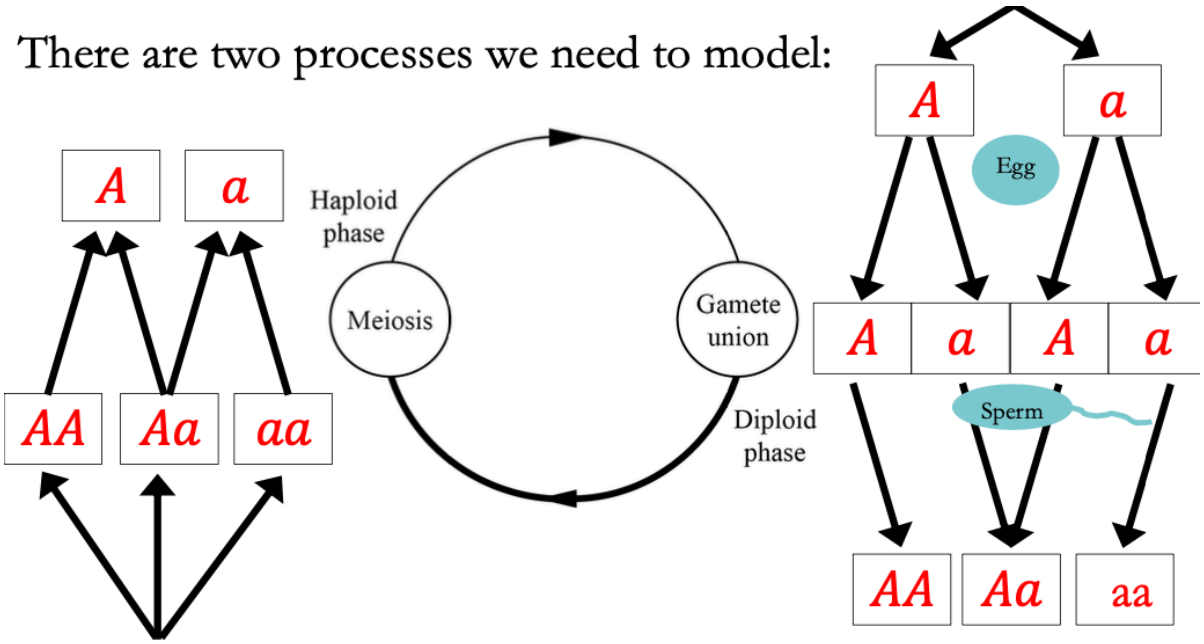
- The change due to selection is the same
- but meiosis breaks apart and reassorts the diploid genotypes.

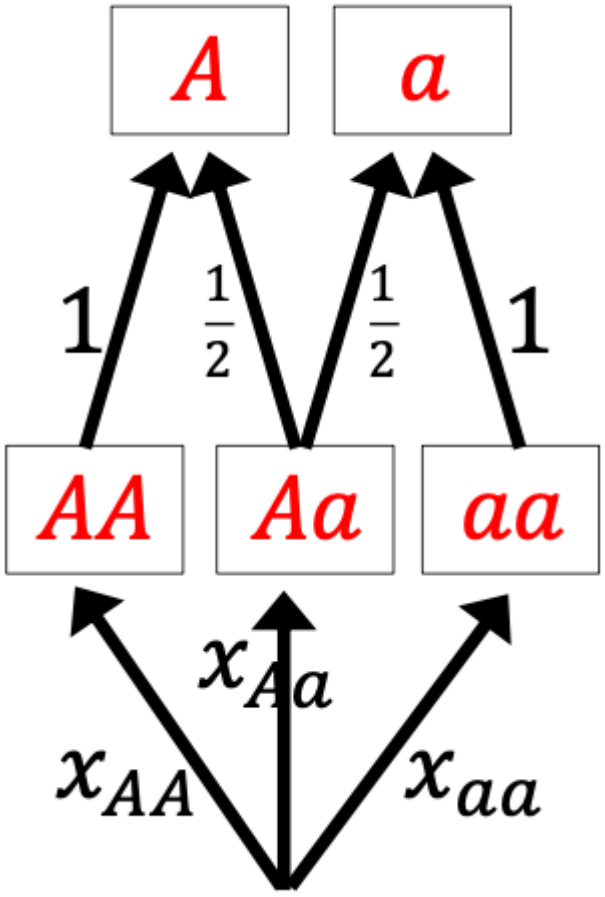
① Sex (segregation) can undo genetic associations built by selection (here between A and a).

Diploid Sexual Life Cycle:

- Part 1: Meiosis segregates alleles to create haploid gametes

There are two processes we need to model:





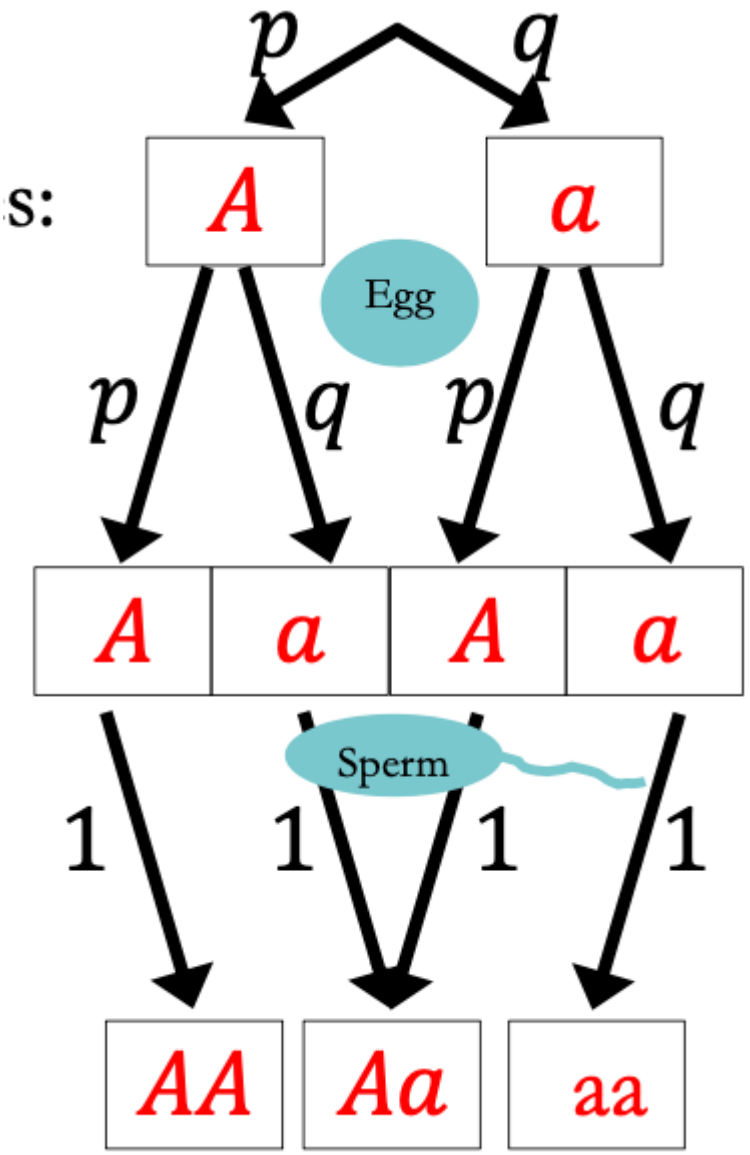
- "Probability tree diagrams" help calculate the chance of different events.

- All options from a node sum to one
- Multiply all probabilities along a path from start to finish to calculate the probability of any one path.

Frequency of A gamete: $p[t+1] = x_{AA}[t] + \frac{1}{2}\, x_{Aa}[t]$

Frequency of a gamete: $q[t+1] = \frac{1}{2}\, x_{Aa}[t] + x_{aa}[t]$

- Part 2: Gamete union brings alleles back together in diploids



Hardy-Weinberg proportions:

$$\text{AA diploid: } x_{AA} = p^2$$

$$\text{Aa diploid: } x_{Aa} = 2pq$$

$$\text{aa diploid: } x_{aa} = q^2$$

D. Diploid Sexuals without Selection

- A Gamete: p[t]
- a gamete: q[t]
    - AA diploid: $x_{AA}[t] = p[t]^2$
    - Aa diploid: $x_{Aa}[t]=2p[t]q[t]$
    - aa diploid: $x_{aa}[t]=q[t]^2$
- Putting together with Meiosis segregation formula

$$\text{Frequency of A gamete: } p[t+1] = x_{AA}[t] + \tfrac{1}{2}\, x_{Aa}[t]$$
$$= p[t]^2 + 1/2\ \ (2p[t]q[t])$$
$$= p[t]\,(p[t] + q[t])$$
$$= p[t]$$

- **Key Points**: With random mating and no selection, allele frequencies do not change and diploids are immediately in Hardy-Weinberg proportions.
- **Assumptions:**
    1. Random combination of gametes from the gamete pool.
    2. No differences in fitness among genotypes
    3. A very large population (no chance effects changing genotype frequencies at any step)

4. No mutation or migration altering variants in population

Example:

- Ritland et al. (2001) in Current Biology found the following genotypes at the melanocortin 1 receptor gene (mc1r) among N=220 sampled bears:

Coat colour	Genotype	Observed
white	GG	22
black	GA	34
black	AA	164

- q = Freq(G) = (22x2+34)/(22x2+34x2+164x2)=0.177, p = 0.823

Observed	Expected	
22	6.91	Nq^2
34	64.2	$N\ 2pq$
164	148.9	Np^2

-

- Low frequency of heterozygotes:
    - Low frequency of heterozygotes
    - assortative mating (preferential mating to the same coat color)
    - Recent immigration of black homozygotes
    - fitness disadvantage of heterozygotes(or combination of these 3)

E. Diploid Sexuals with Selection

① **Basic Formulas:**

- "s" to indicate after selection.

$$\bar{W} = W_{AA}x_{AA}[t] + W_{Aa}x_{Aa}[t] + W_{aa}x_{aa}[t]$$

$$x_{AA}^{s} = \frac{W_{AA}x_{AA}[t]}{\bar{W}}$$

$$x_{Aa}^{s} = \frac{W_{Aa}x_{Aa}[t]}{\bar{W}}$$

$$x_{aa}^{s} = \frac{W_{aa}x_{aa}[t]}{\bar{W}}$$

② **Mix with Segragation**

$$\text{Frequency of A gamete: } p[t+1] = x_{AA}^{s} + \tfrac{1}{2}\, x_{Aa}^{s}$$
$$\text{Frequency of a gamete: } q[t+1] = \tfrac{1}{2}\, x_{Aa}^{s} + x_{aa}^{s}$$

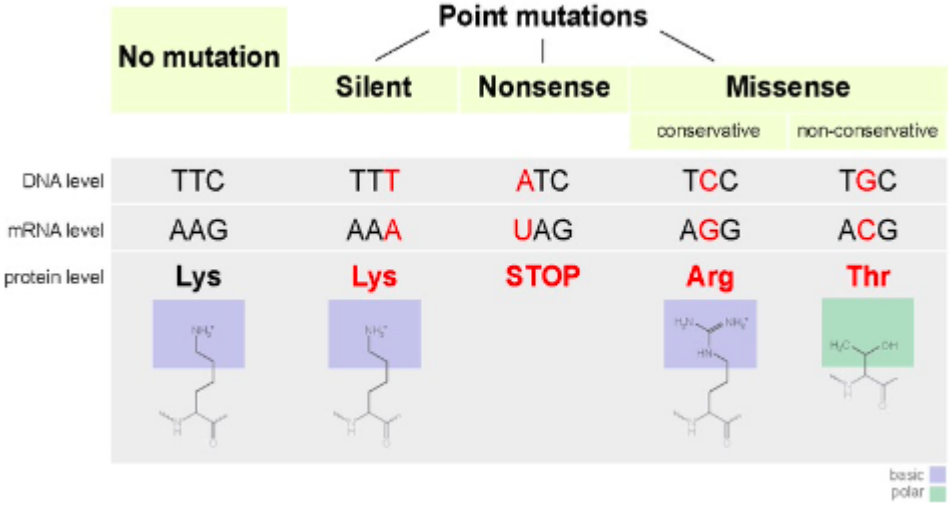
3. MutationPopulations

Mutation: An alteration to the DNA of an organism

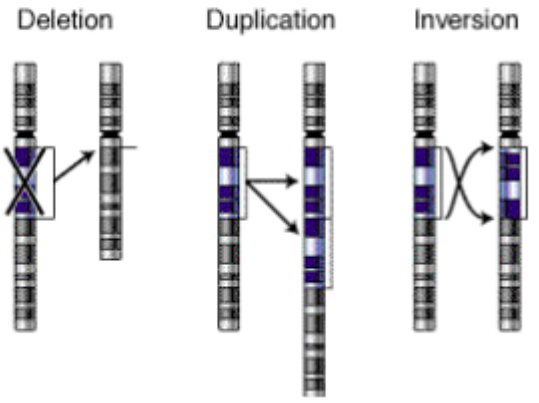
A. Generating novel variation:

① Genetic variance in fitness

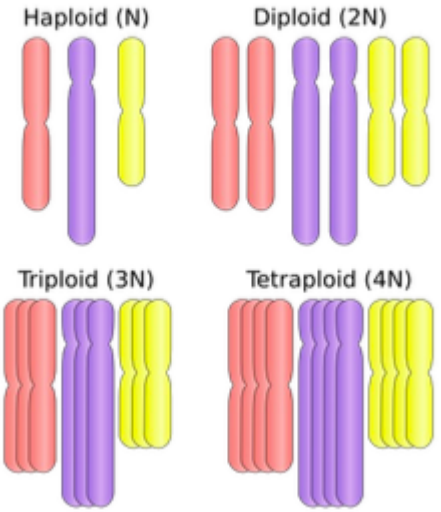
- Differences in the average fitness of alleles at a locus
- The basic fuel upon which evolution acts
- One-Locus Model with Haploid Selection
    - Fixation of A or a (no genetic variation)
- One-Locus Model with Diploid Selection
    - Fixation of A or a (no genetic variation)
    - Unless there is heterozygote advantage (polymorphic equilibrium where alleles A and a are equally fit, on average)



"Structural mutations"



“Genomic mutations”

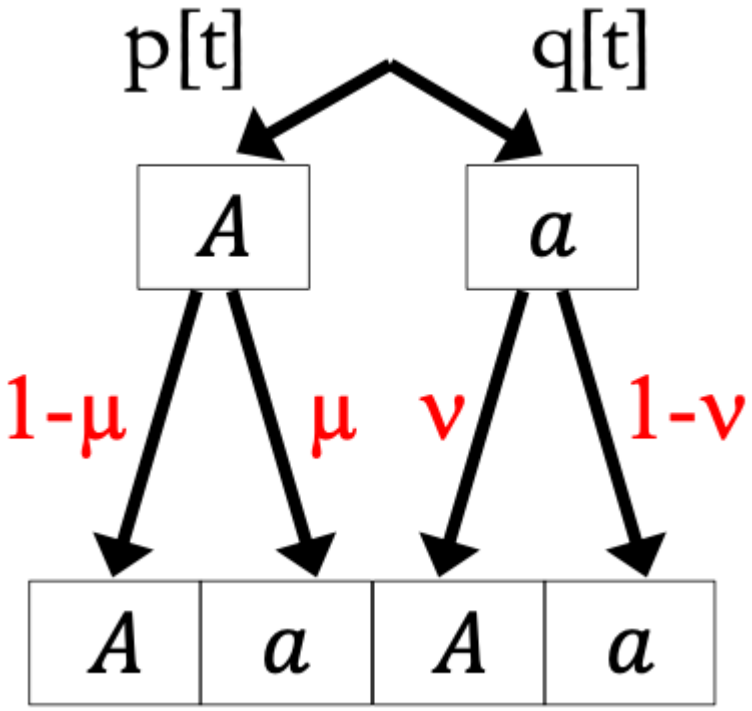


- The error rate in DNA replication per generation varies from organism to organism.
- In eukaryotes, point mutations occur at a rate of roughly 10^{-8} – 10^{-10} per basepair per generation.
    - or roughly 10^{-5} – 10^{-7} per gene per generation
- The error rate also varies from gene to gene
- The mutation rate even depends on the alleles involved e.g., coat color mutations in mice:
    - 11.2 x 10-6 per gene per generation from wildtype to mutant
    - 2.5 x 10-6 per gene per generation from mutant to wildtype
- Mutations disturbing wildtype function (forward mutations, μ) often occur at higher rates than mutations restoring wildtype function (back mutations, v).

B. Mutation without selection

- Mutation rate from A to a: μ
- Mutation rate from a to A: v
- If p[t] is the frequency of A before mutation, then after:
- $p[t+1] = (1-\mu)p[t]+v\ q[t]$

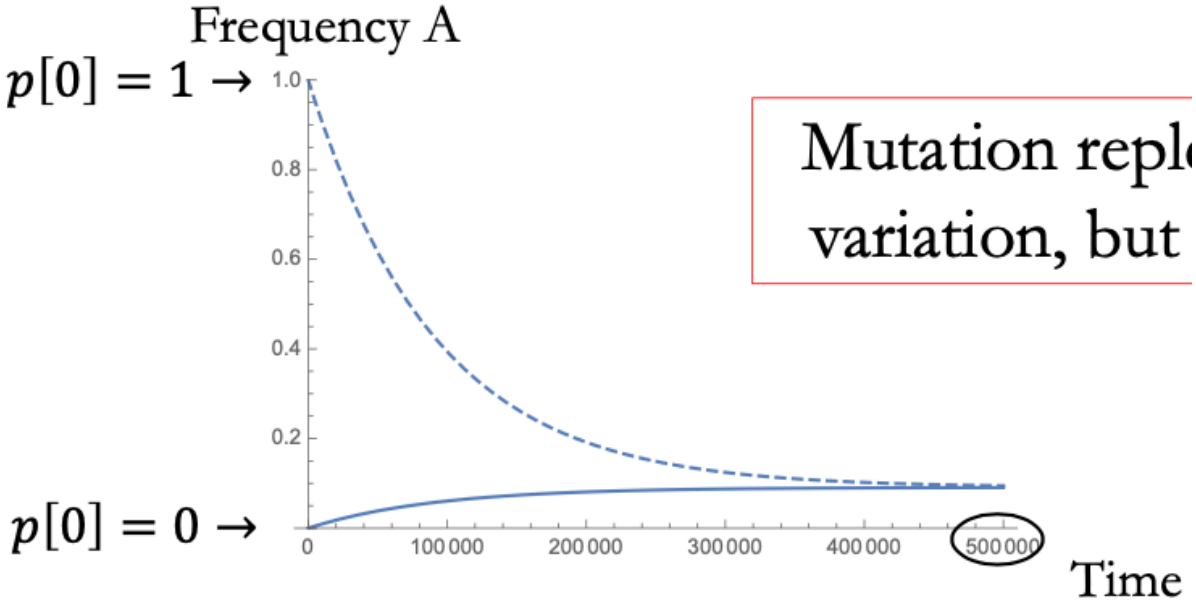
= (1-μ)p[t]+v(1-q[t])



$$p[t+1] = (1-\mu)\,p[t] + \nu\,(1-p[t])$$

$$\hat{p} = \frac{\nu}{\mu+\nu}$$

- With mutation alone, system slowly approaches equilibrium



Mutation repl
variation, but

- While many mutations have no effect on fitness, those that do tend to reduce fitness ("deleterious").
- While selection reduces the frequency of deleterious alleles, mutation continually reintroduces them.

C. Mutation with selection: Haploids

Haploid Fitnesses

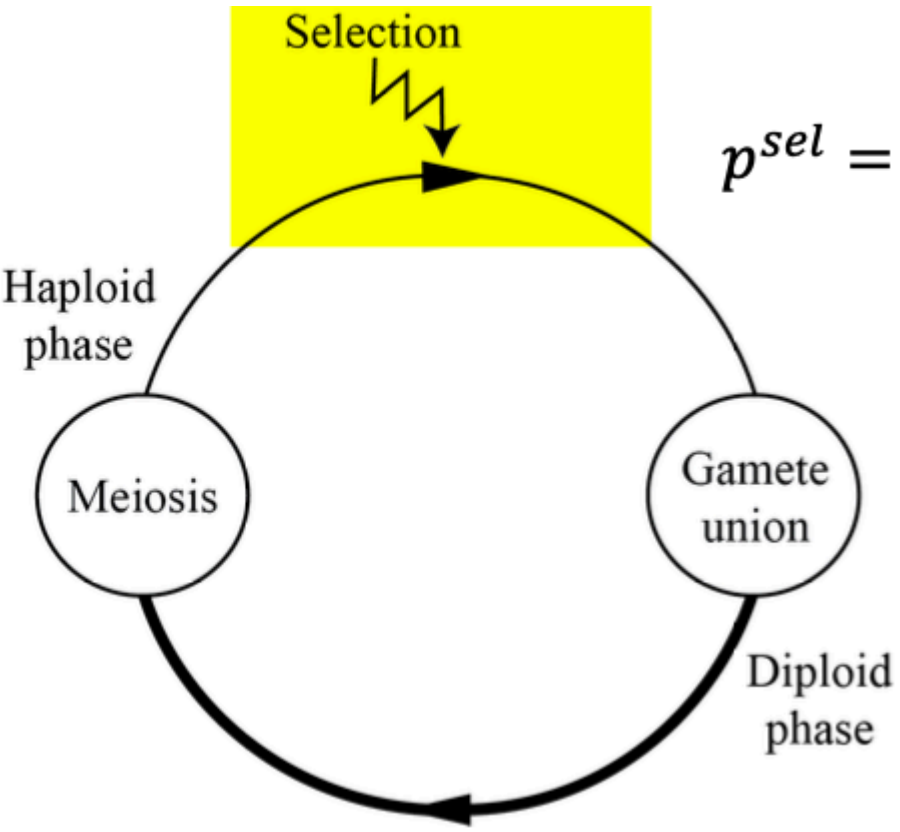
W_A	W_a
1	$1 - s$

- S: selection coefficient
- Selection in haploids: Putting together change due to selection & new mutations

$$p^{sel} = \frac{W_A\, p[t]}{W_A\, p[t] + W_a\, q[t]}$$

$$p[t+1] = (1-\mu)\, p^{sel} + \nu\, q^{sel}$$

$p[t]$


Selection
Haploid phase
Meiosis
Gamete union
Diploid phase

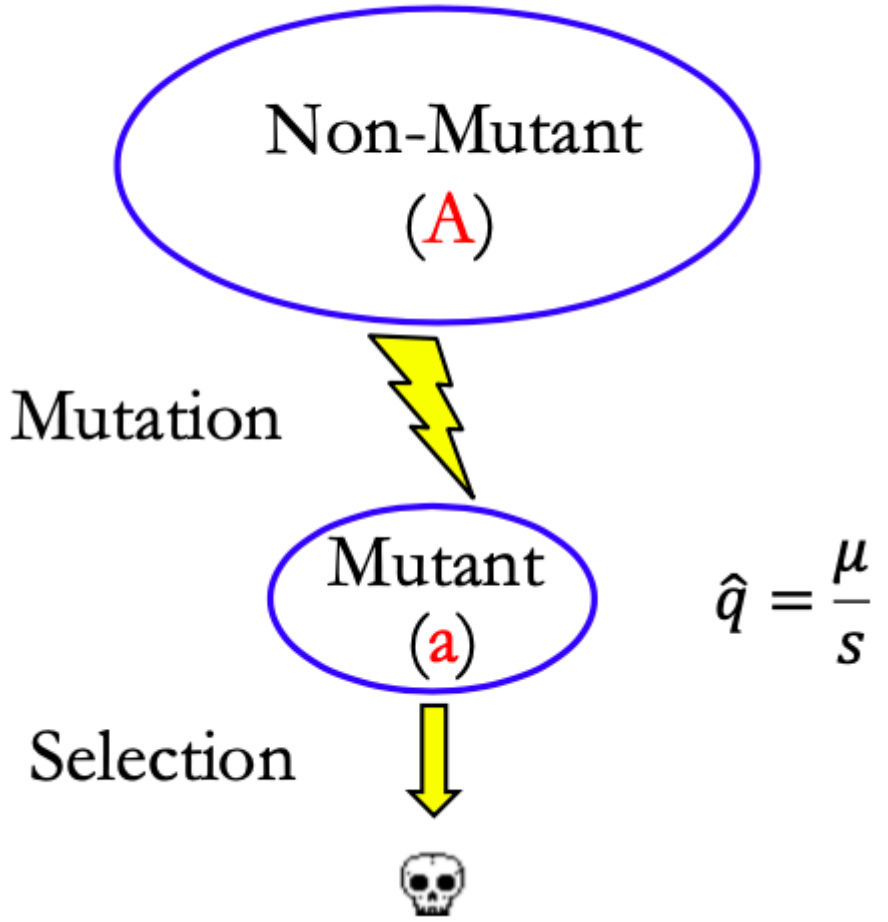
$p^{sel} =$

Equilibrium

$$\hat{p} \approx 1 - \frac{\mu}{s}$$

$$\hat{q} \approx \frac{\mu}{s}$$

Mutation-selection balance



$$\hat{q} = \frac{\mu}{s}$$

- Mutation + selection reach a balance, so harmful alleles persist at a low equilibrium frequency.
- Key point:

- If mutation repeatedly generates deleterious alleles, then they will persist in populations despite their negative effects.
- Selection will act to remove mutant alleles, but mutation will reintroduce them, with these two forces balancing over time.

① Mean fitness:

$$\text{Mean fitness: } \bar{W} = W_A\, p[t] + W_a\, q[t]$$
$$= p[t] + (1 - s)\, q[t]$$
$$= 1 - s\, q[t]$$

Mutatio

- Key point: With mutation and selection, mean fitness can decline over evolutionary time
- For example, starting with a population that is entirely non-mutant A, mean fitness will decline from 1 as new mutations arise ($q[t]$ increases)
- Key point: With mutation and selection, mean fitness can decline over evolutionary time

"Mutation load":

- Reduction in fitness due to mutations = μ (sometimes called fraction of "selective deaths")
- Key point: Mutations reduce fitness on average by an amount that depends only on the mutation rate

Mean fitness at equilibrium:

$$\bar{W} = 1 - s\,\hat{q} = \mathbf{1} - \boldsymbol{\mu}$$

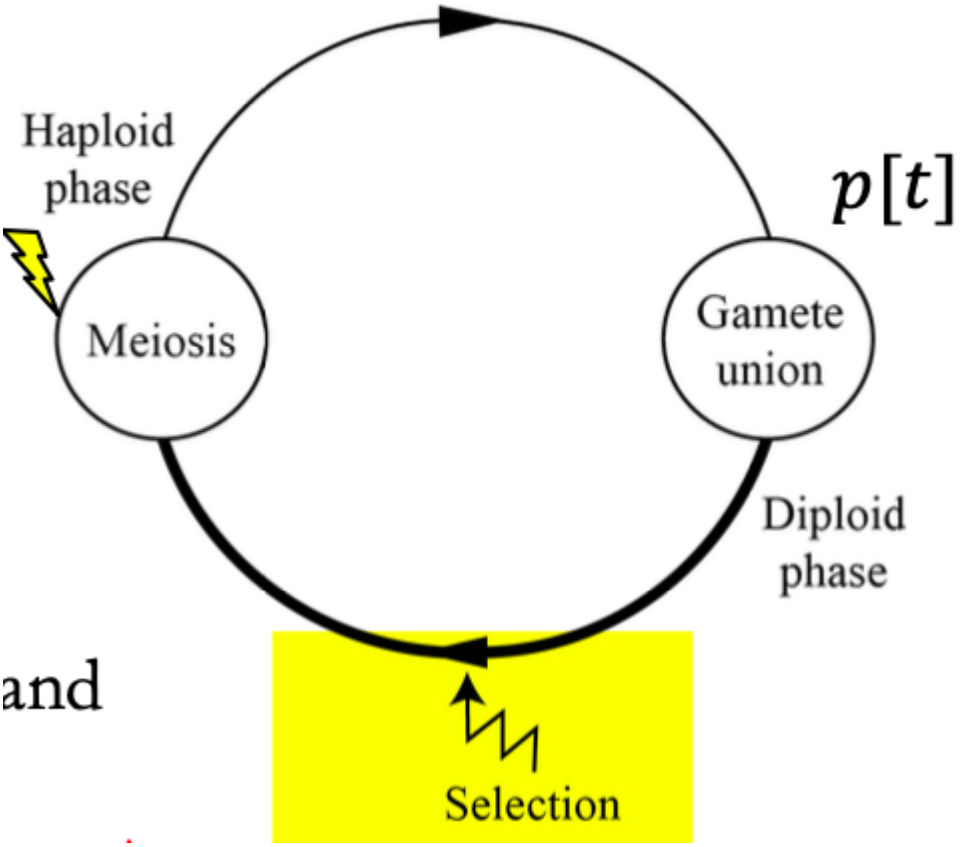
- More severe mutations equilibrate at a lower frequency & less severe mutations at a higher frequency, but the number of deaths will depend only on the mutation rate.

D. Mutation with selection: Diploids

Diploid Fitnesses

W_{AA}	W_{Aa}	W_{aa}
1	$1 - hs$	$1 - s$

- s is the selection coefficient and
- h is the dominance coefficient



$$p^{sel} = \frac{W_{AA}\, p[t]^2 + W_{Aa}\, p[t]q[t]}{W_{AA}\, p[t]^2 + W_{Aa}\, 2\, p[t]q[t] + W_{aa}\, q[t]^2}$$

$$p[t+1] = (1-\mu)\,p^{sel} + \nu\,q^{sel}$$

Not recessive ($h>0$)	Recessive ($h=0$)
Equilibrium	Equilibrium
$\hat{p} \approx 1 - \frac{\mu}{hs}$	$\hat{p} \approx 1 - \sqrt{\frac{\mu}{s}}$
$\hat{q} \approx \frac{\mu}{hs}$	$\hat{q} \approx \sqrt{\frac{\mu}{s}}$

the equilibrium (shown on Canvas) where $\hat{p} = p[t] = p[t+1]$

Key point: Mutant alleles are expected to be found at higher frequency when recessive ($\hat{q} \approx \sqrt{\frac{\mu}{s}}$) than when they affect heterozygous fitness ($\hat{q} \approx \frac{\mu}{hs}$).

- Assuming ***h*** >0, the mean fitness at equilibrium: **1 − 2*μ***
- Mutation load: Reduction in fitness due to mutations = 2μ

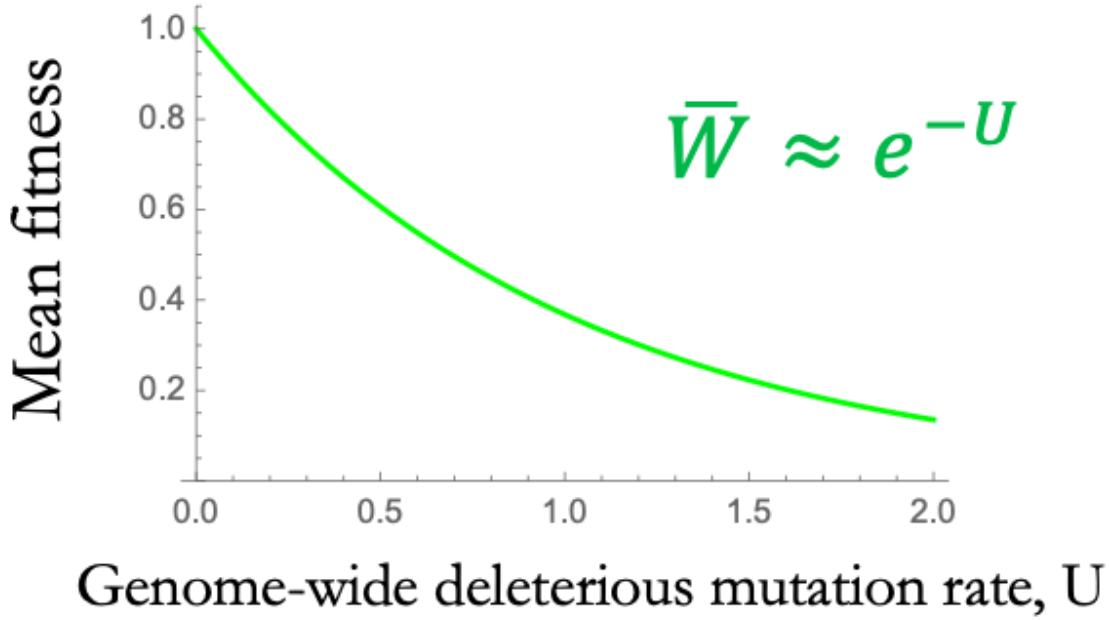
$$\bar{W} = \hat{p}^2(1) + 2\hat{p}\hat{q}(1-hs) + \hat{q}^2(1-s) \approx 1 - 2\mu$$



- For example, starting with a population that is entirely non-mutant AA, mean fitness will decline from 1 to 1 − 2μ.
- **If there are L loci, the total reduction in fitness is:**

$$\bar{W} = (1-2\mu)^L \approx e^{-2\mu L} = e^{-U}$$

- where U is the sum-total deleterious mutation rate across the diploid genome per genome.



E. Case Study

① Albinism is a recessive condition that occurs in humans at a frequency of 1/20,000. If individuals with albinism have a relative fitness of 0.9, what is the mutation rate to the albinism allele?

Equilibrium (recessive)

$$\hat{q} \approx \sqrt{\frac{\mu}{s}}$$

- h=0 (recessive)
- s=0.1 (relative fitness of 0.9)

- Affected individuals (aa) occur at frequency q^2

$$\hat{q} \approx \sqrt{\frac{1}{20000}}$$

- μ=0.1/20000x1=5x10^{-6}

② Achondroplasia is a partially dominant condition affecting bone growth, and is the most common cause of dwarfism.

- Homozygosity at this locus is lethal.
- Affected individuals are heterozygous and historically had a relative fitness of 0.2.
- The phenotype occurs at a frequency of 1/10,000 in the human population.

Estimate the mutation rate to the achondroplasia allele?

Equilibrium (not recessive)

$$\hat{q} \approx \frac{\mu}{hs}$$

- $1 - hs = 0.2$

$$1/10{,}000 \approx 2\hat{p}\hat{q} \approx 2\hat{q}$$

- since $\hat{p}$ is near one and homozygotes are rare

-μ= 4x10^{-5}

===Mutation load (reduction in mean fitness due to mutations) equals the mutation rate (2μ in diploids).

5: Genetic drift: the role of chance

Genetic drift : the change in allele frequencies that results from random processes that take place within populations over generations.

Also known as “random genetic drift” or just “drift”

A. Diploid Model without Selection

A gamete: $p[t+1] = p[t]$

① Assumption: With random mating and no selection, allele frequencies do not change and diploids are immediately in Hardy- Weinberg proportions.

1. Randomcombinationofgametesfromthegametepool.
2. Nodifferencesinperformanceamonggenotypes
3. Consider smaller population (allowing chance events that change genotype frequencies)
4. No mutation or migration altering variants in population

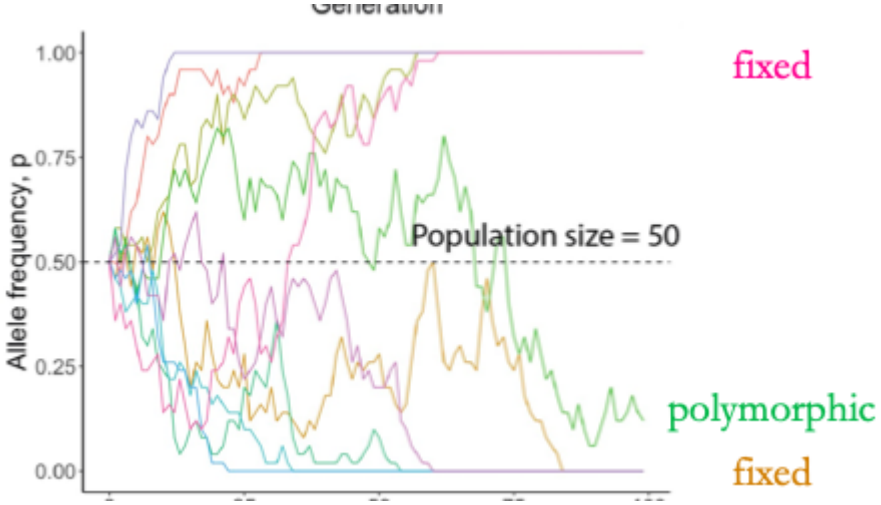
② When do random events take place?

- A sample of adults contributes to the gamete pool
- Mutations occur randomly
- A sample of juveniles survives to become adults
- A sample of gametes is combined to make zygotes
- A sample of zygotes survives to become juveniles

- Because all populations are finite, drift is always acting,
- producing small or large changes in allele frequencies and affecting loci throughout the genome.
- Like chance events in our lives, drift acts to bring good luck and bad luck to the evolutionary process.

③ predictable effects

- Changes in allele frequencies from one generation to the next are expected to be greater in smaller populations.
- Populations remain polymorphic for longer, on average, with increasing population size.



- The effects of drift are driven mainly by population size, therefore, all else being equal, drift acts similarly across all of the loci in the genome.
- Sampling affects all of the loci carried by an individual... chance acts on good, bad and neutral alleles.
- And even the loss of bad or neutral alleles means loss of genetic variation that might become beneficial later.

B. How does drift work?

- If there are n copies of allele A and N–n copies of allele a in a population of size N,
    - the probability that allele A eventually fixes is p=n/N
- On average, the amount of time it takes for a single allele to fix is twice the number of alleles within the population:
    - →2N generations with N haploid individuals
    - →4N generations with N diploid individuals (2N alleles)

- The same is true if we look backwards in time and ask how long ago did all of the alleles descend from the same single ancestral allele (Coalescence time):
    - →2N generations with N haploid individuals
    - →4N generations with N diploid individuals

①expected heterozygosity

- The loss of genetic variability due to drift can be measured by calculating the "expected heterozygosity" (H[t]) of a population over time, t.
    - "Expected heterozygosity" (H[t]) = The probability that two alleles drawn at random are different alleles.
    - For example, if we started with a randomly mated population at time t: H[t] = 2 p[t] q[t]
- When we sample two alleles at time t, there are two possibilities:
    - Sampled alleles descended from the same parent allele: Probability = 1/2N
    - Sampled alleles descend from different parent alleles: Probability = (1– 1/2N)
- Ignoring mutation, the only way for the alleles to differ at time t is if they descended from different parent alleles, which occurs with probability 1-1/(2N).

$$H[t] = \underbrace{\left(1-\frac{1}{2N}\right)}_{\text{Two gene copies descended from different parent genes}} \underbrace{H[t-1]}_{\text{\& those parent genes were different}}$$

- Key Point: Genetic variability declines, on average, by 1/(2N) every generation in a population of N diploids.

②expected homozygosity

expected homozygosity: F[t] = 1–H[t]

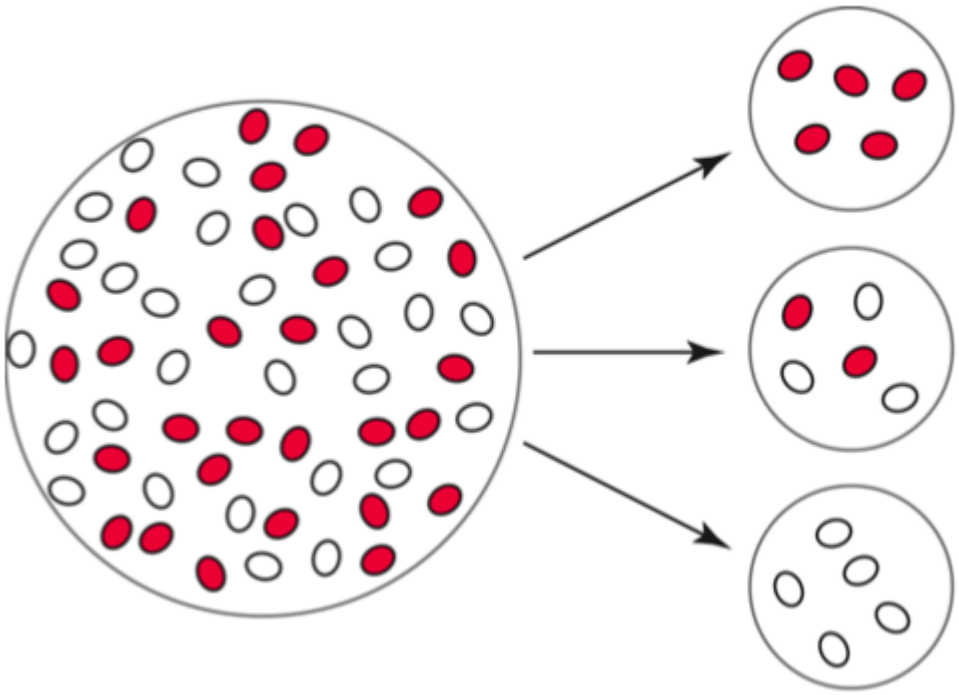
$$F[t] = 1 - H[t] = 1 - \left(1 - \frac{1}{2N}\right) H[t-1] = 1 - \left(1 - \frac{1}{2N}\right)(1 - F[t-1])$$

$$= \frac{1}{2N} + \left(1 - \frac{1}{2N}\right) F[t-1]$$

- Describes the increase in homozygosity when alleles have recently descended from the same ancestor in a finite population (“inbreeding”).
- Small populations, in particular, become “inbred” with little genetic variation remaining due to drift.

C. Founder effects

- Definition: effects that occur when a new population is founded by a small number of individuals from a larger source population

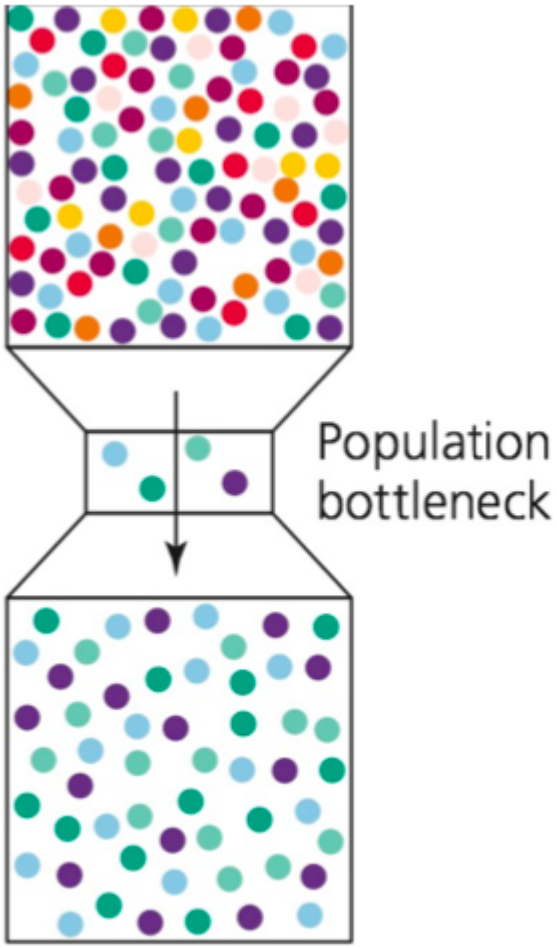


- Case Study: Population history of French Canadians

- Many French Canadians descend from ~8500 French settlers (arriving from 1608-1759)
- ~30 disease-causing alleles and genetic disorders are especially common in this population.
    - Cystic Fibrosis
    - People with cystic fibrosis are homozygous for mutations in CFTR (recessive disease).
    - Individuals with one defective copy (heterozygotes) are called "carriers".
    - Carriers chance is 1/15 in French Canadian

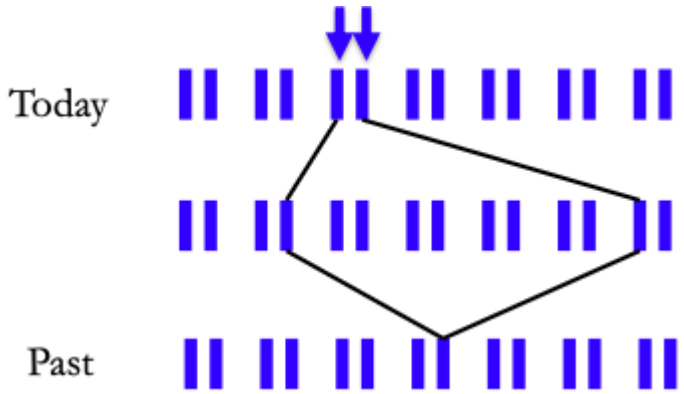
D. bottleneck effect

- A severe reduction in the number of individuals in a population, which results in the loss of genetic variation in the surviving population.



- Example:
    - Black-footed Ferret: Only 7 of the 18 individuals brought into captivity had surviving offspring

① Inbreeding: With few surviving lineages, the chance that the two alleles in an individual recently shared a common ancestor is much higher (= “inbreeding”), increasing the frequency of homozygous recessive diseases (= “inbreeding depression”).



- While we have focused on neutral alleles (no selection), selection is always intertwined with drift.
- That said, if selection is strong enough and the population size large enough, the impact of drift is less noticeable.

③ probability of fixing

- Haldane (1927) showed that in a large population, the probability of fixing a new mutation is twice its selective advantage when rare
- 2s in haploids, 2hs in diploids
- The fate of an allele is determined more by selection than drift in diploid populations when 2hs > 1/(2N), Haldane's result with selection> Fixation probability for a neutral allele
- Kimura (1962) generalized this result showing that the probability of fixation of an allele at initial frequency p is*:

$$\frac{1-e^{-2cNsp}}{1-e^{-2cNs}}$$

- *The c refers to ploidy level:
  - Haploids: c = 1 with WA=1+s, Wa=1,
  - Diploids: c = 2 with WAA=1+2s, WAa=1+s, Waa=1 (formula assumes additivity)

- Key points: Even slightly deleterious mutations can fix due to drift. Beneficial mutations are often lost!

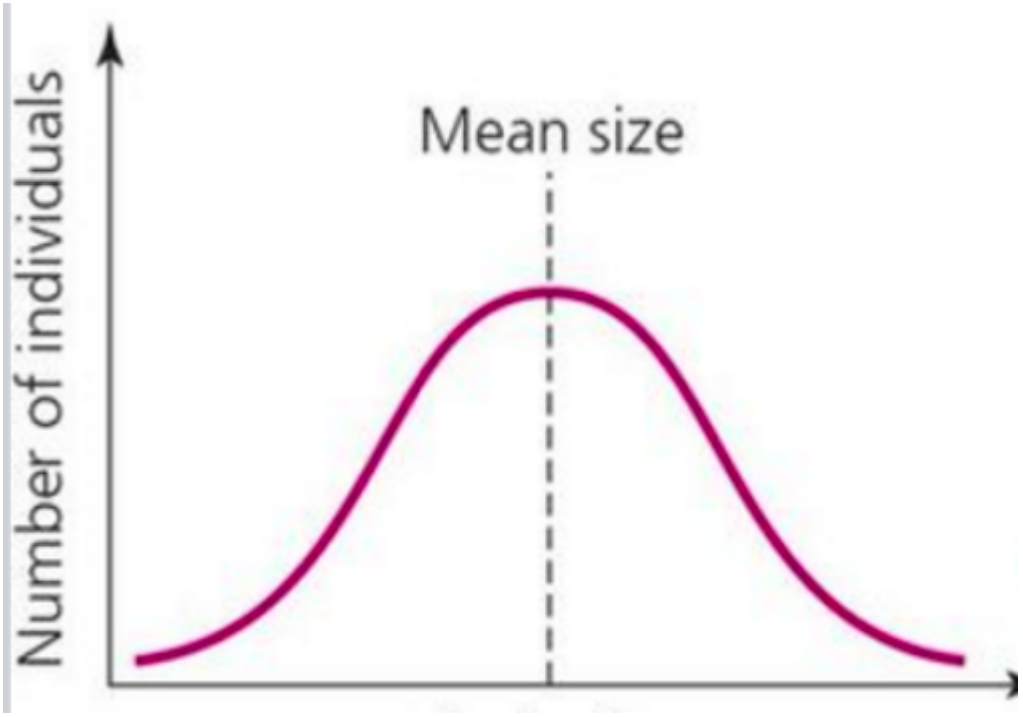
Part III. Adoption and Traits

1. Quantitative trait evolution

A. Quantitative Traits:

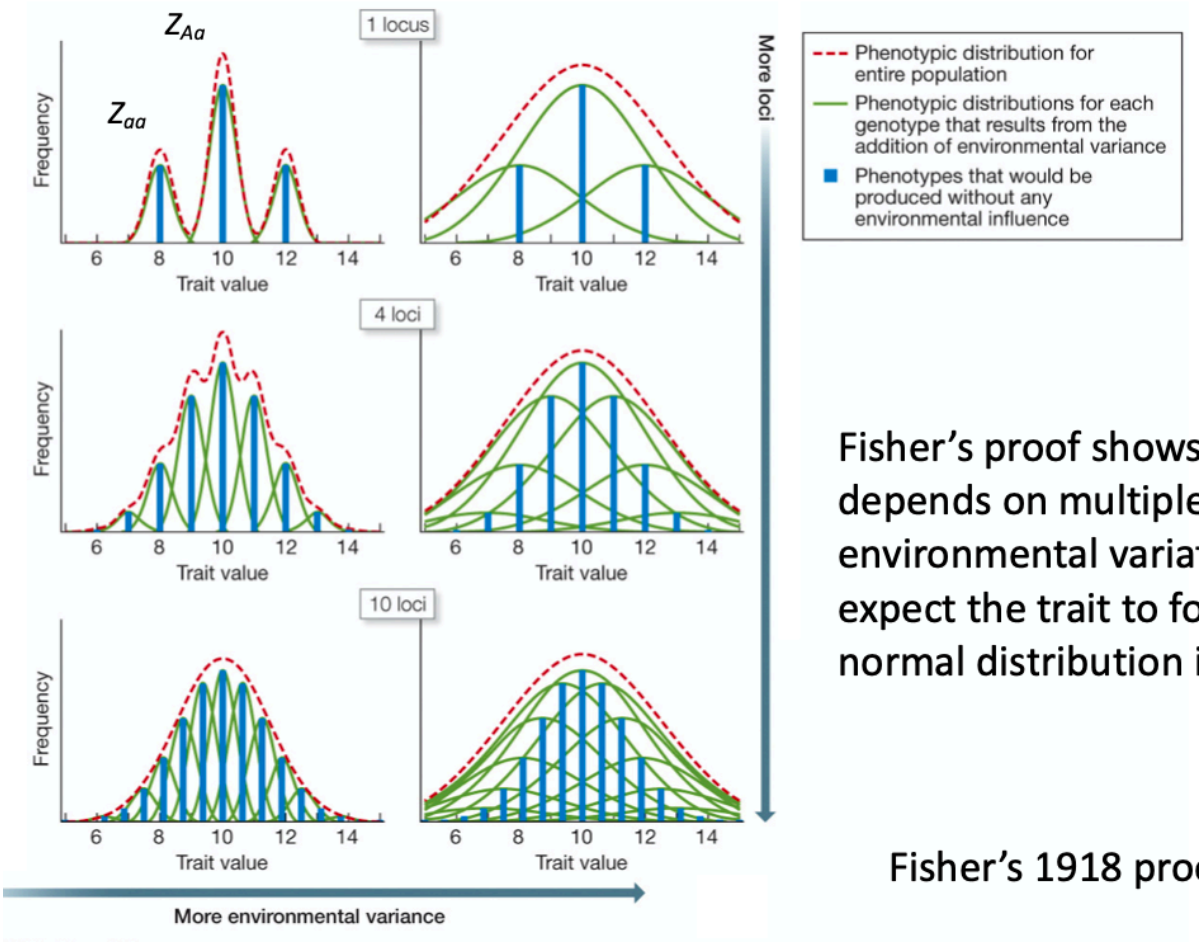
① Concepts:

- Traits with no discrete categories of phenotypes,
- instead a pattern of continuous trait variation
- best described with a mean and a variance.



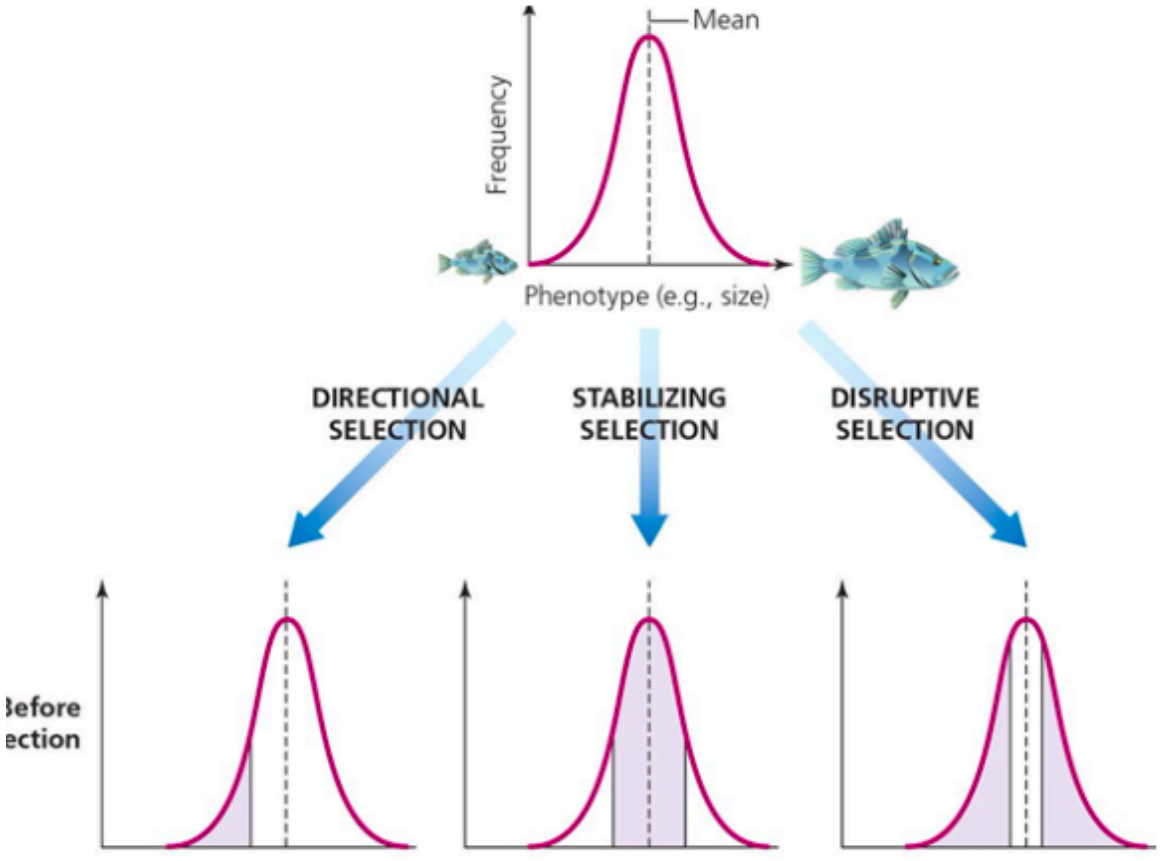
② Fisher's 1918 Proof of many loci:

- if a trait depends on multiple loci and environmental variation
- then the trait follows a normal distribution in nature



③ Selection on Quantitative Traits

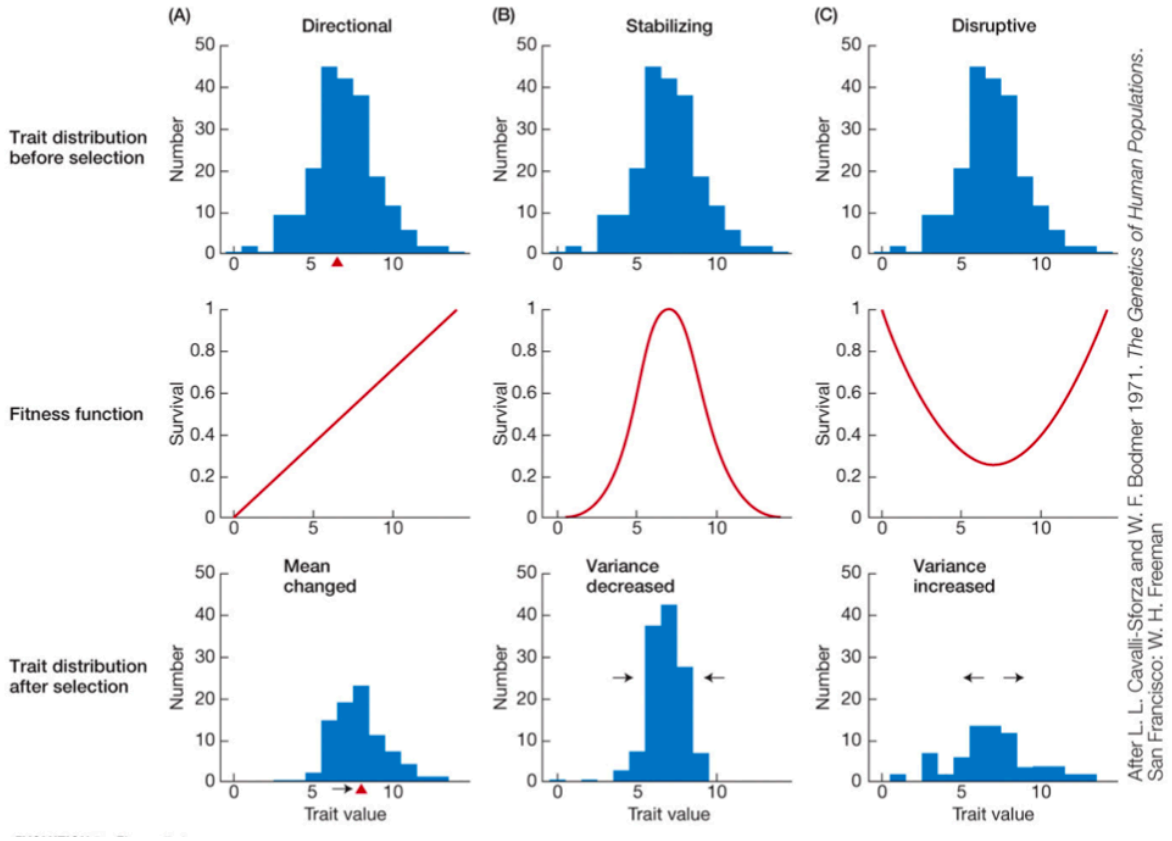
- Selection is now acting on a distribution, not just a single value
- Three Forms of Selection on Quantitative Traits:



Directional Selection

Stabilizing Selection

Disruptive Selection



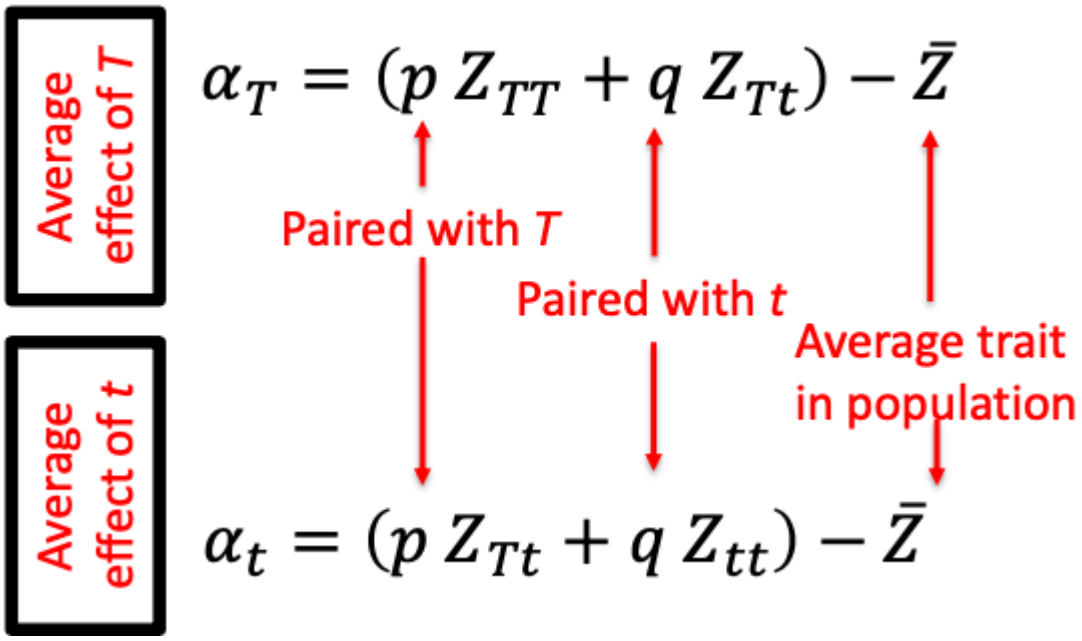
- Each form of selection has a predicted impact on patterns of trait variation seen after selection, within the same generation.

B. Inheritance of quantitative traits

- We will focus on diploids:
    - Key point: A parent cannot pass their trait phenotype on to their offspring, they only pass along half of that offspring's genes*.
    - TT parents with Z_{TT} phenotype (Z=trait value for that particular genotype) pass only one T allele and may produce either TT homozygous offspring or Tt heterozygous offspring.

① average effect

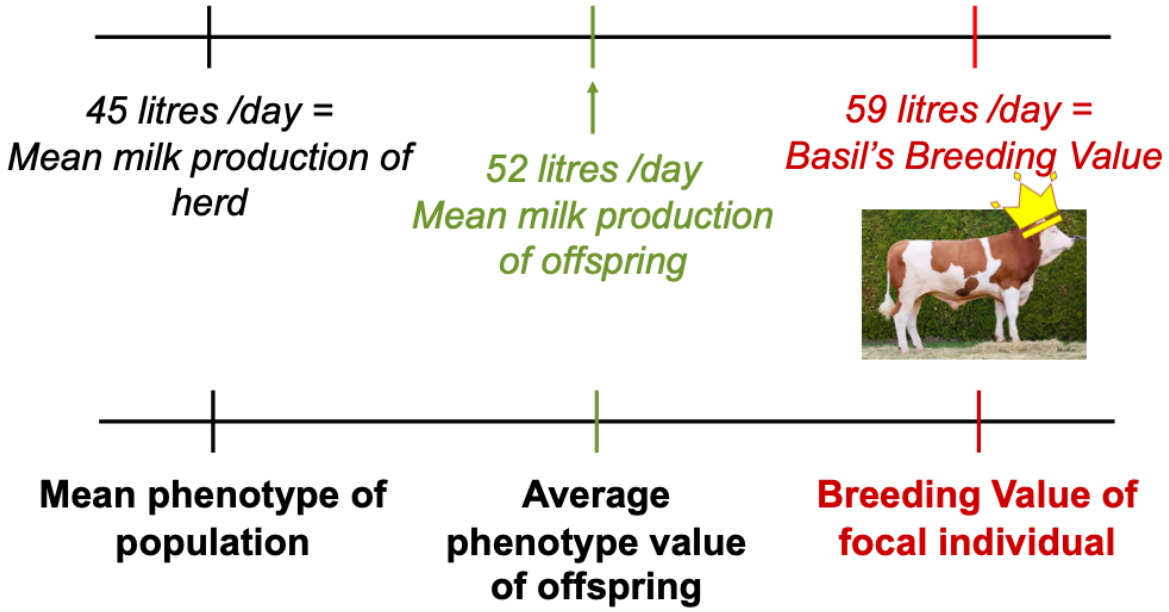
- At one gene, the "average effect" of an allele determines how much the mean of a trait ($\overline{Z}$) changes when we know that an individual carries this particular allele (assume random pairing)



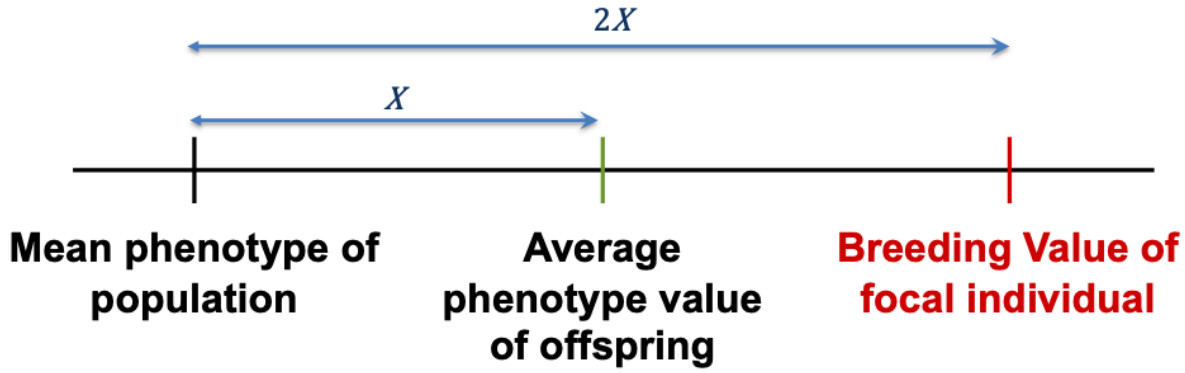
② breeding value（A）

- The breeding value of an individual (or genotype) measures the average contribution of that individual to the trait mean of its offspring, if crossed with an average range of mating partners.
- *Question to solve*:
    - Which individuals should be picked to reproduce?
    - If we choose an individual to cross, what will the offspring be like?
    - How much will the offspring resemble this parent?
- Case Study:
    - If milk production is heritable, then a cow with high milk production is a good candidate for a breeding program.
    - *How would we select the bulls for our breeding program? "Mint" or "Basil"*
    - Each bull is mated with a set of females, who average 45 litres of milk per day; the milk production of their daughters is measured.
    - Basil's daughters produce an an average of 47 litres of average of 52 litres of milk

- Mint's daughters produce milk per day per day.
- Conclusion: Basil has a higher breeding value than mint



- The breeding value of an individual = 2 x (population mean trait- its offsprings' mean trait)
- "2x" because the breeding value sums the average effects of both alleles at all loci in the diploid parent



- Key points:
  - The breeding value is not a measure of the phenotype of an individual
  - The breeding value measures the degree to which an individual's phenotype is expected to be transmitted to that individual's offspring.

③ Relationship between the breeding value, A, and phenotype, P, of an individual

- The overall phenotype, Phenotype, of an individual is a function of its Genetics and Environment ($P = G + E$)
- The genetic component (G) can be further broken down into:
    - A = Additive genetic effects (“Breeding” value)
    - D = Dominance effects, and
    - I = Interaction effects (epistasis between loci)
- As these are measured as effects relative to the mean, they sum to give:
    - The Genotypeas $G=A+D+I$
    - The Phenotypeas $P=(A+D+I)+E$
- DEFINITIONS
- Genetics (G):
    - Additive effects of alleles (A): The average effect of all alleles that contribute to a phenotype (also known as the breeding value)
    - Dominance effects (D): The effects of dominance interactions among alleles at each locus
    - Epistasis (I): The effect of interactions among alleles at different loci
- Environment (E): Includes all of the non-genetic influences on the phenotype (nutrition, resources, parental care, stressors, etc.)
- Why additive effects ?
    - Non-additive effects(D,I)are disrupted every generation, and therefore do not have a predictable effect on the offspring.
    - Environmental effects (E) are (by definition) not genetically based.
    - These three types of variation affect the phenotypic trait values (P) but are not predictably passed on to offspring.

- It’s the additive genetic effects that natural selection is going to act on.

C. Inheritance in Populations: Heritability

- Breeders of plants and animals focus on picking individuals for crosses in order to try to move the mean trait values and/or change the traits of their group (a cultivar/herd). (Artificial selection)
    - Eg coastal dog
- In nature, selection on differential success of individuals with different trait values affects how much a trait's mean of a population will change in response to selection. (Natural selection)
    - Eg coastal wolf

① The phenotypic variance (V_p)

- Concept: the variability among a set of individuals for a particular trait.
    - Not all of this variation is due to genetic differences (V_G),
    - Not all of the genetic differences among individuals are passed on to the next generation (V_A).
- Components:
    - V_p : total phenotypic variance
    - V_G : total genetic variance
    - V_A: genetic variance due to additive effects of alleles, called the **Additive Genetic Variance**
    - V_D: genetic variance due to dominance among alleles
    - V_I: genetic variance due to epistatic interactions of alleles at different loci
    - V_E: Phenotypic variance due to environmental effects
- These components of the phenotypic variance sum together:
    - These components of the phenotypic variance sum together:

- $V_p = V_G + V_E$,in which,
    - $V_G = V_A + V_D + V_I$, and therefore,
- $V_p = V_A + V_D + V_I + V_E$

② Additive genetic variance

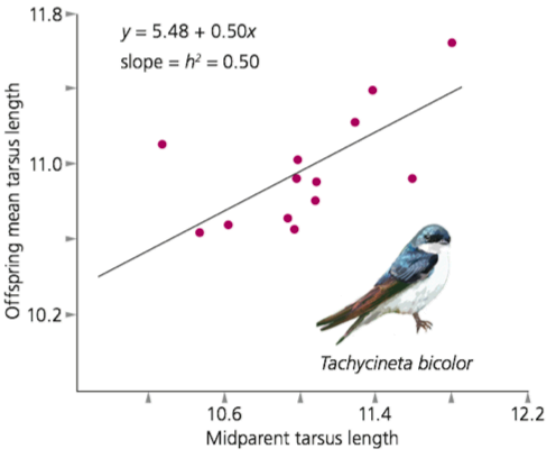
- V_A describes the average degree to which offspring are expected to resemble their parents; it equals the genetic variance due to additive effects of alleles and is also equal to the variance in breeding values.
- Thus, V_A is key to understanding how much of the observed phenotypic variation is available to respond to selection (meaning to shift the population mean in the next generation).
- VA is also key to estimating the heritability of traits.

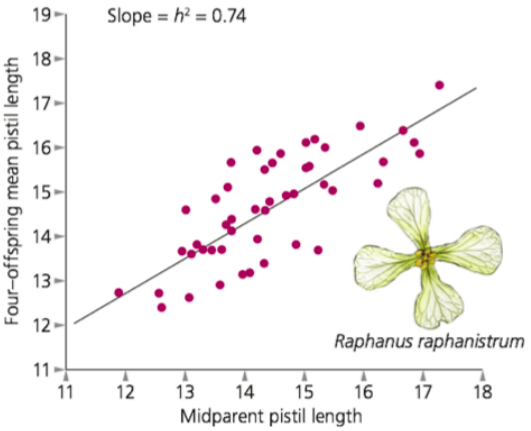
③ Heritability of a quantitative trait

- The inheritance of a quantitative trait is measured by the proportion of phenotypic variance that is caused by additive genetic variance (the "narrow sense heritability", h^2).
    - $h^2 = V_A / V_p$
    - $h^2 = V_A / (V_A + V_D + V_I + V_E)$
    - Heritability describes the extent to which we expect trait variation in the parental generation to be passed to offspring.
- because V_A is a fractional component of V_p , heritability ranges from 0 to 1.

④ Parent-offspring Regression to Estimate Heritability

- Heritability: a measure of the expected similarity between parents and offspring.
- In the plot of value of the offspring(x) against the trait score of the parents(y), the slope of the line is the heritability, h^2.





- Note: midparent value = average of the phenotypic value of the parents
- Midoffspring value: mean phenotypic value of the offspring

D. Three Important points about heritability

① Estimates of heritability depend on the *population* and *environments* in which they were estimated.

$$h^2 = \frac{V_A}{V_P} = \frac{V_A}{V_A + V_D + V_I + V_E}$$

- 1). V_E (and therefore V_P) will differ in different environments.
  - The impact of the environment on heritability estimates depends on the trait and can be substantial.
- 2). V_G will vary among populations, and also within populations over time.
  - Genetic variance shifts as allele frequencies change.
  - When V_G changes, V_A ,V_D ,V_I also change, and therefore heritability changes.

② Heritability estimates are for populations, not individuals.

- The value of h^2 does **not** predict specific trait values of individual offspring
- It is a measure of the average relationship of traits values in parents and offspring across the population as a whole.

- Eg. Knowing the heritability estimates for body mass does not allow us to predict the specific mass of offspring of a pair of parents of known body mass.

③ Heritability does not indicate the degree to which a trait is genetically based.

- Rather, it measures the proportion of the phenotypic variance that is the result of (additive) genetic factors.
- For example, a trait may be under genetic control, but if there is no additive genetic variance associated with the trait, the heritability estimate will be 0.

④ Narrow-Sense Heritability and Effects of Selection

- h^2 = how much offspring resemble their parents.
- Selection acts on phenotypes (fitness differences).
- Only the heritable part of those differences can show up in the next generation.
- Therefore: h^2 determines how much selection changes the trait mean.
- Higher h^2 → stronger evolutionary response; lower h^2 → weaker response.

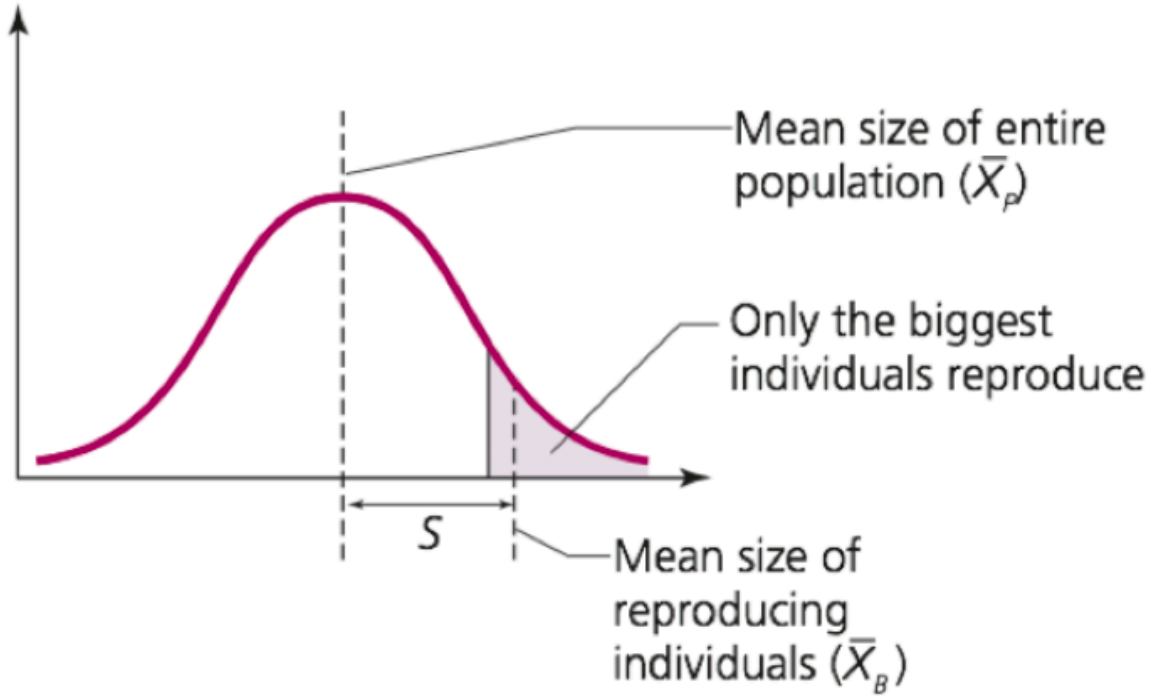
E. The breeder's equation

$$R = h^2 S \quad h^2 = R/S$$

- how much we expect a trait mean to change (respond) over generations, given the heritability of a trait, and knowledge of the strength of selection acting on that trait.
- S = selection differential
    - = how much the selected parents differ from the population mean (strength of selection within one generation).
- R = response to selection
    - = how much the trait mean shifts in the next generation.

① The response to selection

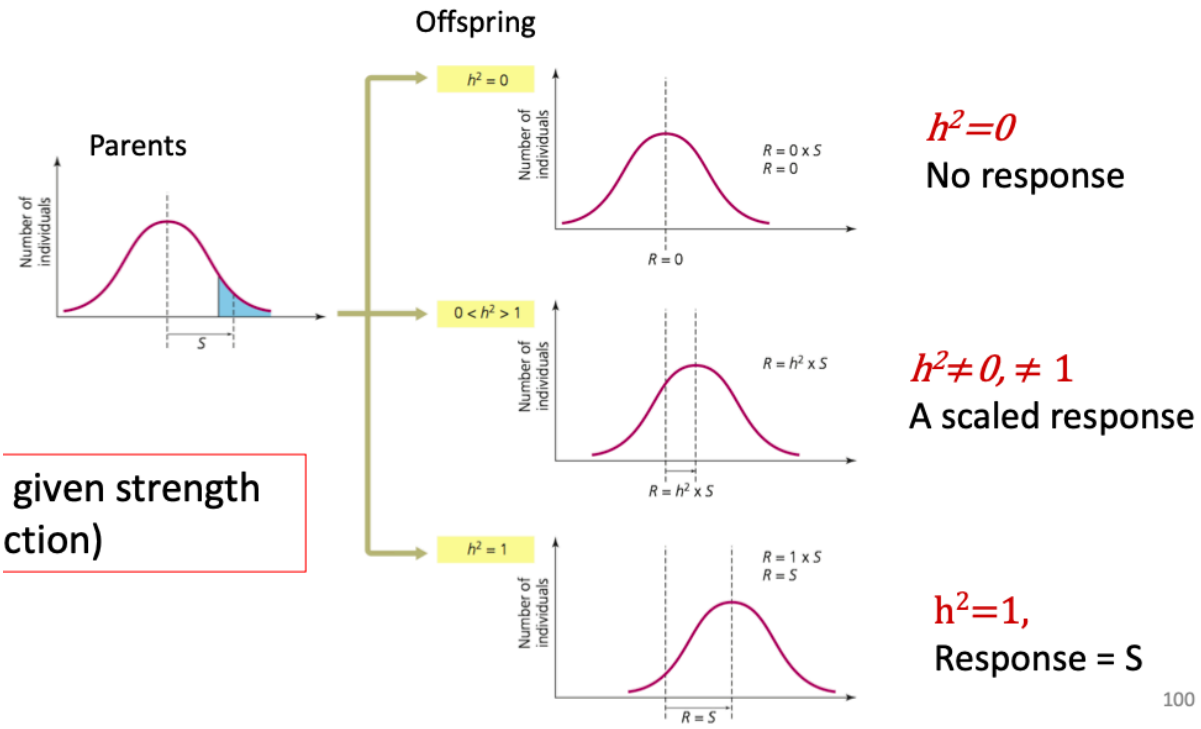
- What do we need in order to predict how much the mean trait value would shift under directional selection?
    1. The starting mean trait value*
    2. The heritability of the trait
    3. The strength of selection



S = Selection Differential

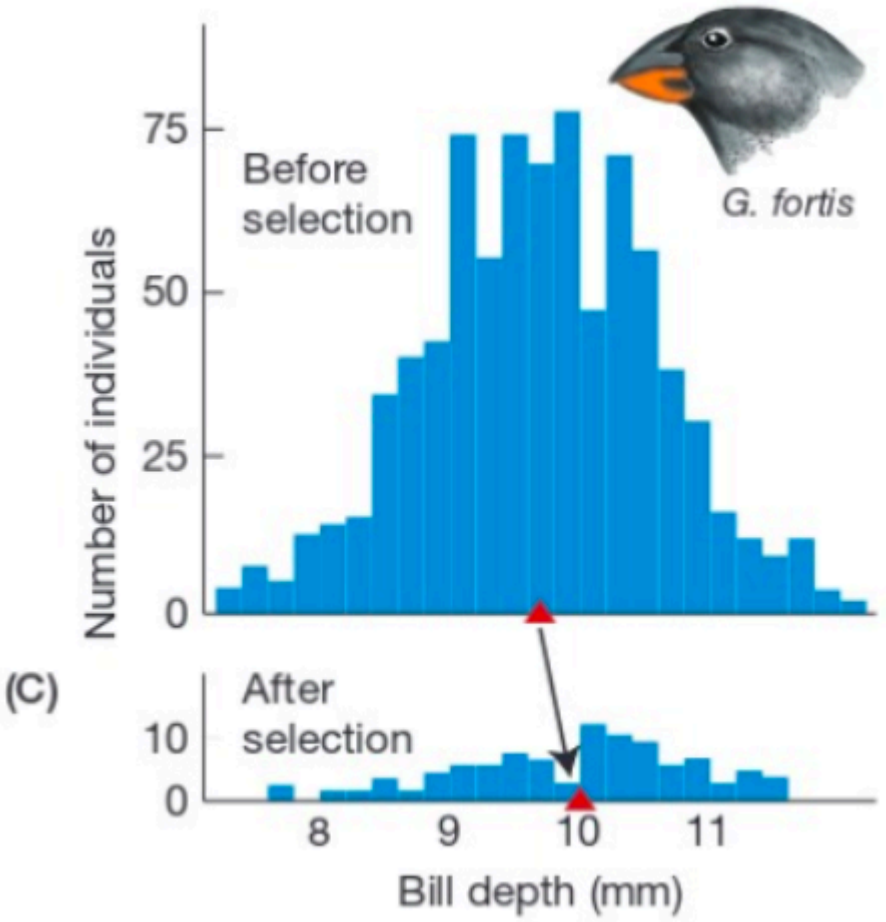
$$S = \bar{X}_B - \bar{X}_P$$

② Higher heritability translates to a greater response *(in the offspring generation

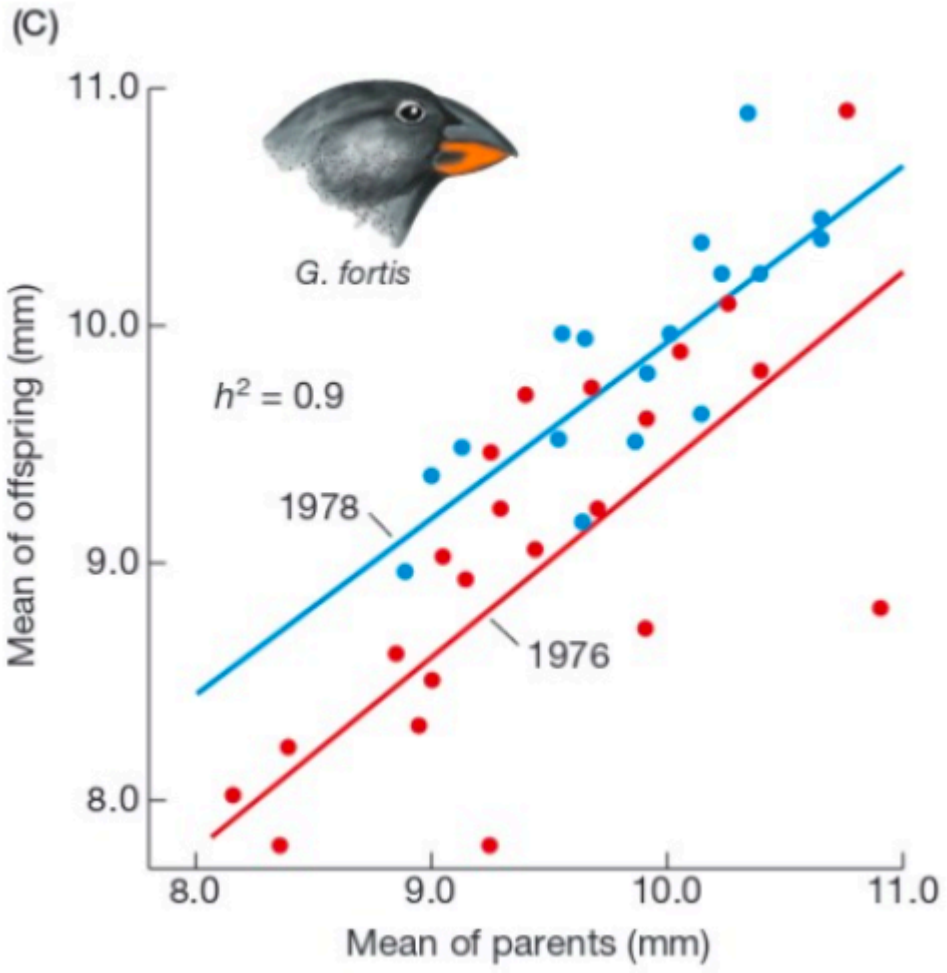


③ Case Study 1: Geospiza fortis beak depth

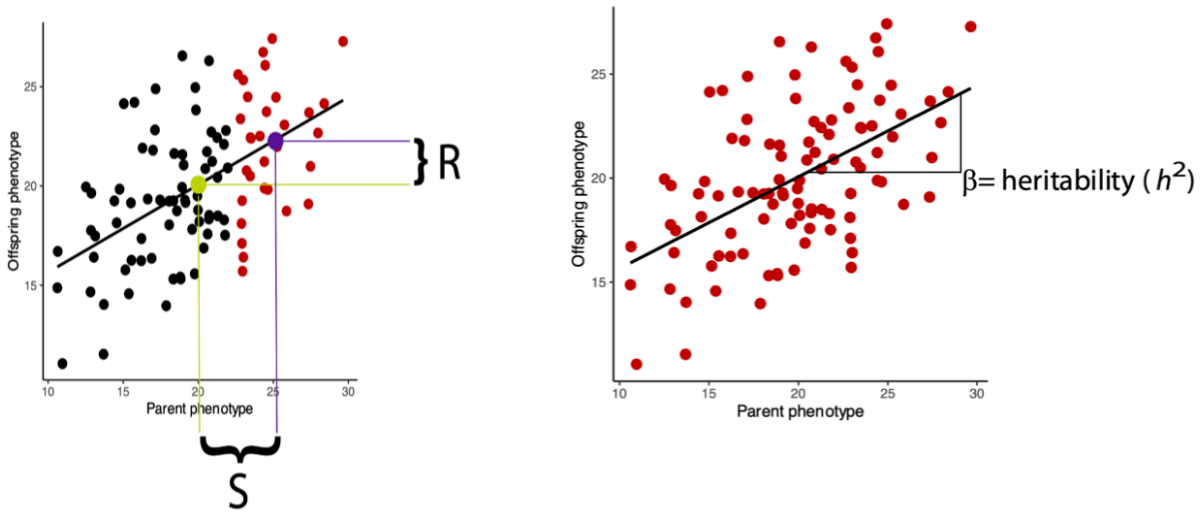
- **1977 drought** caused major food shortage; small/soft seeds disappeared.
- Birds with **deeper, stronger beaks** survived better → directional selection.
- **Mean beak depth before drought (1976): 9.4 mm (N = 751).**
- **Mean beak depth of survivors (1978): 10.1 mm (N = 90)**



- **Selection differential:** S=10.1−9.4=0.7 mm.
- Parent–offspring regression gave **high heritability: $h^2 \approx 0.9$**.



- Offspring of survivors also had **larger beaks** → mean trait increased
- **Conclusion:** Natural selection + high heritability → **rapid evolutionary change** in beak depth.
- $R=h^2 S=0.9\times0.7=0.63$ mm
- Average X of offspring=9.4+0.63=10.03 mm



④ Case Study 2: Fungi in Heat

A population of Cordyceps has mean tolerance of 28.6 C. Climate change leads to a heat dome, with the average tolerance of survivors being 31.3 C.

(a) What is the selection differential? S = 31.3-28.6=2.7

(b) If thermal tolerance has a heritability of 0.3, what is the expected response to selection? R = 0.3*2.7=0.81

(c) What is the predicted mean desiccation tolerance of the population in the next generation? 28.6+0.81=29.41

⑤ Case Study 3— Non-additive genetic variance example

Setup

- A plant's height is determined by a single locus with **overdominance**
- Allele frequencies
    - $A_1 = 0.5$
    - $A_2 = 0.5$
- Genotypes, frequencies, and heights:
    - $\mathbf{A_1A_1}$: freq 0.25, height **10 cm**
    - $\mathbf{A_1A_2}$: freq 0.50, height **12 cm** (highest)
    - $\mathbf{A_2A_2}$: freq 0.25, height **10 cm**

Mean height before selection

- $\bar{z}=0.25(10)+0.50(12)+0.25(10)=11$ cm

Selection

- Strong selection removes all plants **< 11 cm**.
- Survivors = **only A_1A_2** (height = 12 cm).
- Mean height of selected parents = **12 cm**.

Selection differential

$S=12-11=1$ cm

Key outcome: no evolutionary change

- Survivors are all A_1A_2.
- They mate randomly:

$$A_1A_2 \times A_1A_2 \rightarrow \begin{cases} 1/4\ A_1A_1\ (10\text{ cm}) \\ 1/2\ A_1A_2\ (12\text{ cm}) \\ 1/4\ A_2A_2\ (10\text{ cm}) \end{cases}$$

- This produces **exactly the same genotype frequencies as the original population**.

Mean of offspring: $\bar{z}_{offspring}=11$ cm

Conclusion:

- **No evolutionary change (R = 0)** even though **S = 1 cm**.
- Reason: **$V_A = 0$** because heterozygote advantage (overdominance) produces **non-additive genetic variance only**, not additive variance
- With **$h^2 = 0$**, the Breeder's Equation predicts:

  $R=h^2S=0$
- **Strong selection does not guarantee evolution.**
  - **If $V_a = 0$ (e.g., overdominance), allele frequencies do not change → no evolutionary response.**

2. Human Origin

A. History of the Theory:

- The Great Debate
- Scopes ‘monkey’ Trial, 1925
    - Tennessee had an act of law (Butler Act) that prohibited the teaching of evolution
    - John T Scopes (a high school teacher) violated the act by teaching evolution
    - This precipitated the “trial of the century”
    - The resolution: evolution can be taught but the human part

B. What makes humans special?

1. Language
2. Bipedalism
3. Intelligence
4. Tool use
5. Fire

C. Human fossils

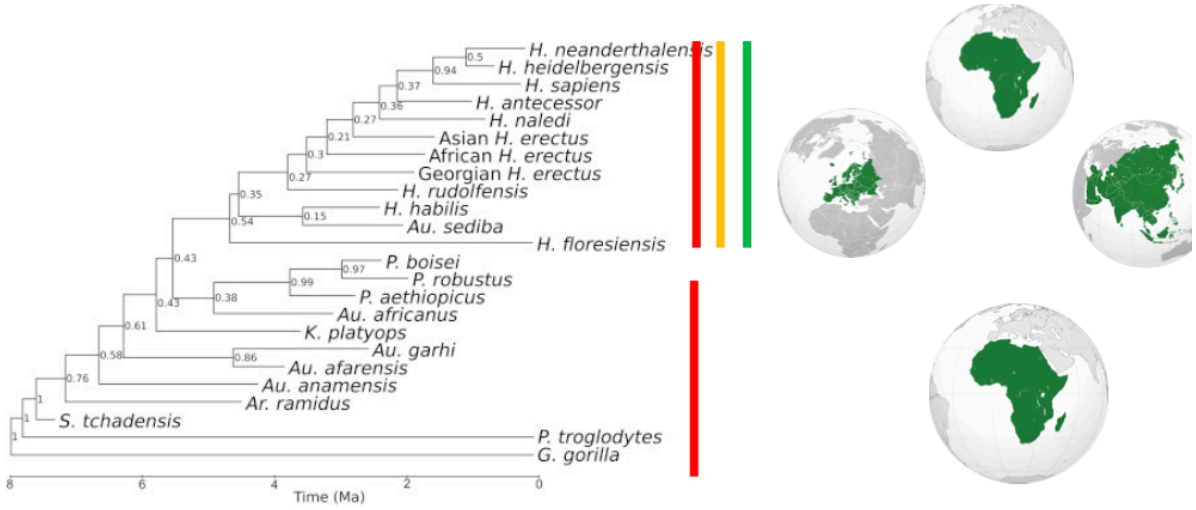
①fossil problem

- Many species known from just a few fragments
    - High-dimensional measurement
    - Phylogenetics
    - Shared trait – in this case, teeth





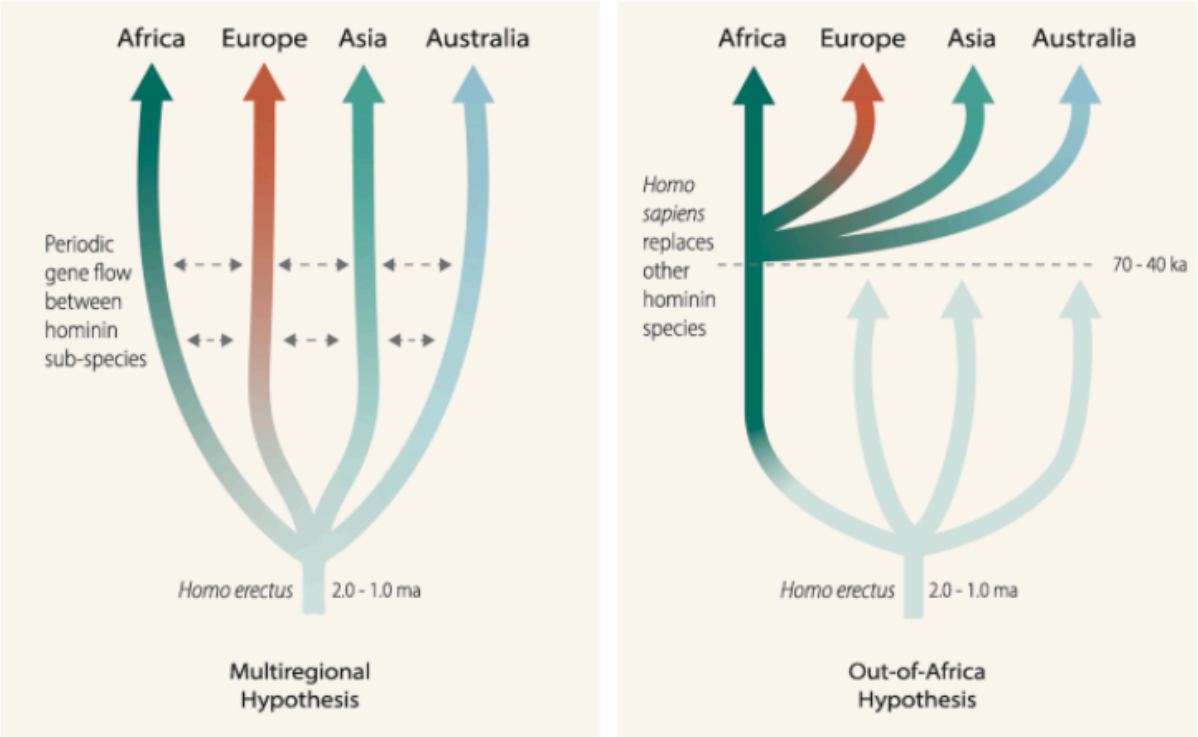

- “Absence of evidence”
    - Fossil evidence is unambiguous – the genus Homo originated in Africa



- Picture changes
    - The traits that make us human are inherited from our ancestors
    - Many species of human walked the Earth at the same time
    - Early Hominids were in Africa, later Homo species were global

D. Out-of-Africa Debate

- How modern Homo sapiens are related to the Homo erectus populations that existed throughout Africa and Asia was long the subject of debate.
- Two ideas: multiregional or out-of-Africa.



- The multiregional hypothesis: modern Homo sapiens evolved from other Homo spp in Africa and Asia, with gene flow keeping populations similar
- The out-of-Africa hypothesis: modern Homo sapiens evolved recently within Africa, then migrated to Asia and Europe

E. Genetics

①Mitochondrial Evidence

- Cann et al. 1987 — 147 human mtDNA sequences
- mtDNA maternally inherited → traces female lineage only
- mtDNA phylogeny estimated MRCA ≈ 200,000 years ago
- This ancestor = “Mitochondrial Eve”

Key patterns

- Human mtDNA sequences very similar globally → recent common ancestry
- African individuals show highest mtDNA diversity
- Non-African regions have subset of African mtDNA lineages

Conclusion

- Supports Out-of-Africa model
- Modern Homo sapiens evolved in Africa; migrations exported only part of African diversity.

②Y chromosome(male ancestry.)

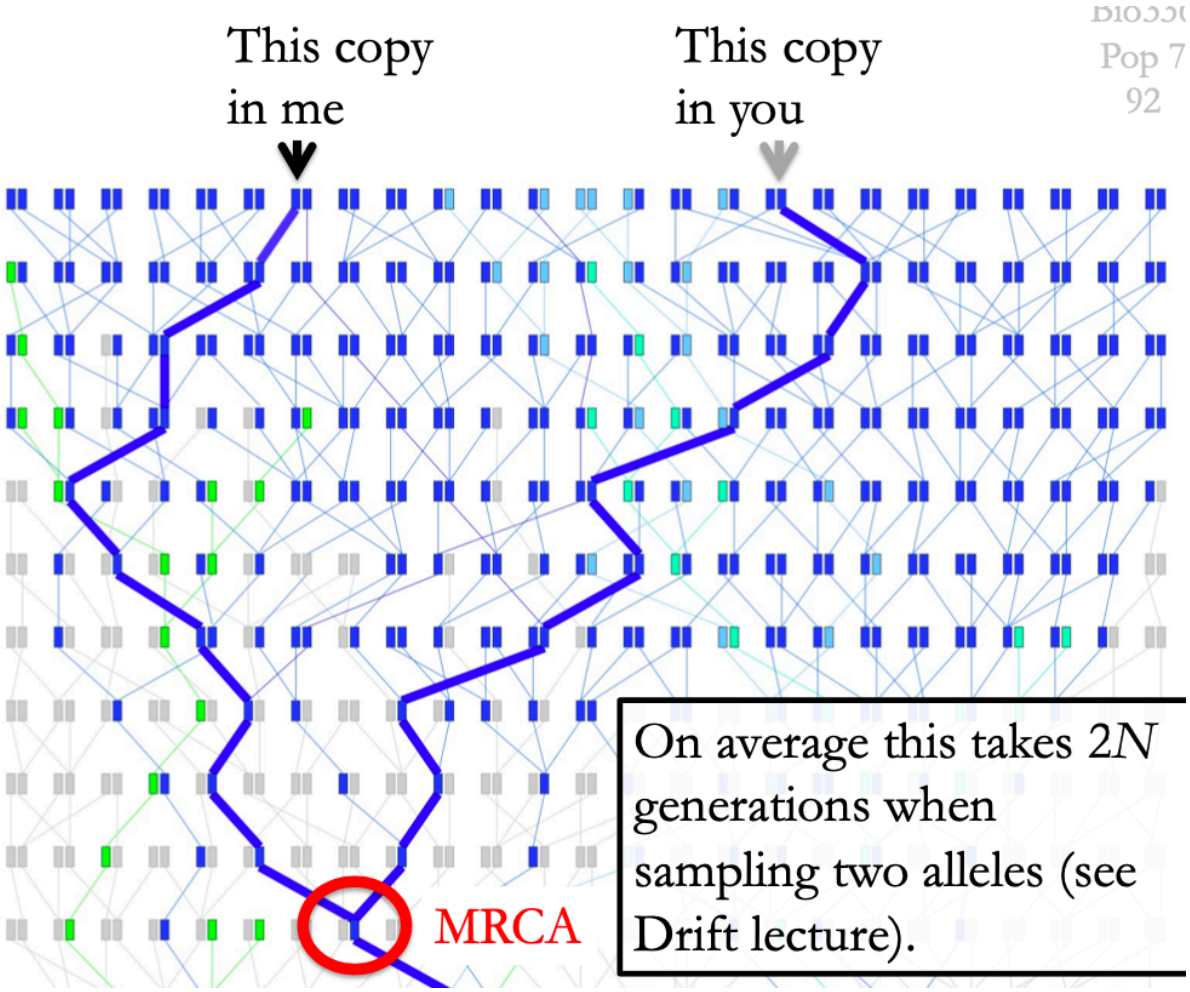
- The Y-chromosome MRCA ("Y Adam") lived ~50,000 years ago.
- All modern Y chromosomes are closely related → recent common ancestor.
- Different Y lineages are common in different regions → reflects migration history.
- One widespread Y lineage (~0.5% of males worldwide) coalesces ~1,000 years ago, likely due to rapid expansion from a single historical male ("Genghis Khan" hypothesis).
- Genetic similarities show global migration routes.
- Humans reached the Americas very recently, during the last Glacial Maximum (via the Bering land bridge).

Genetic evidence supports:

- Homo sapiens evolved in Africa and later migrated globally ("Out of Africa").

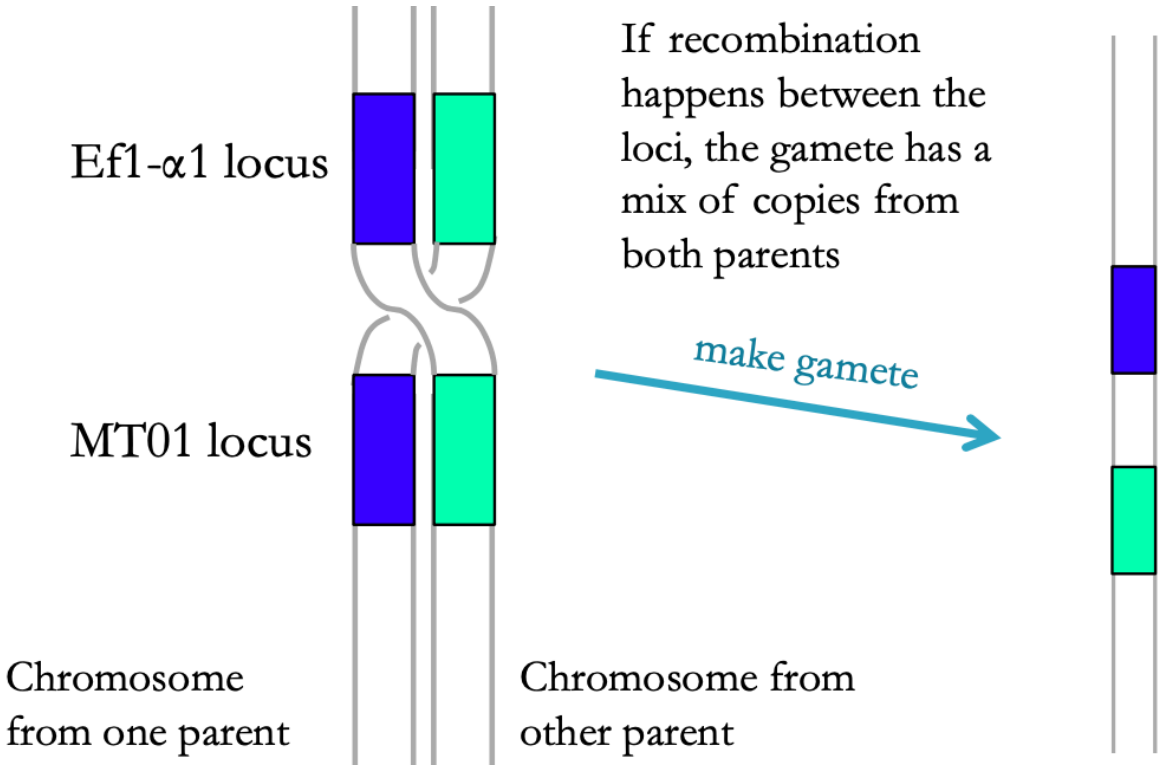
F. Recombination shatters the genome into different trees.

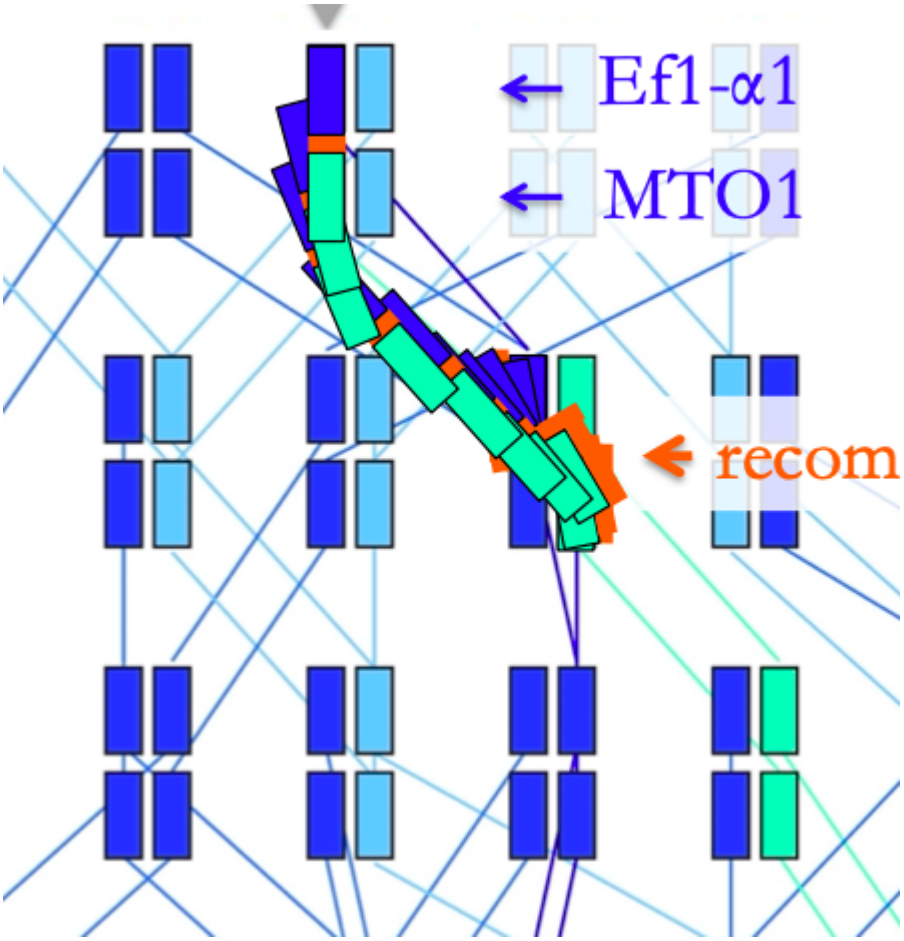
① MRCA

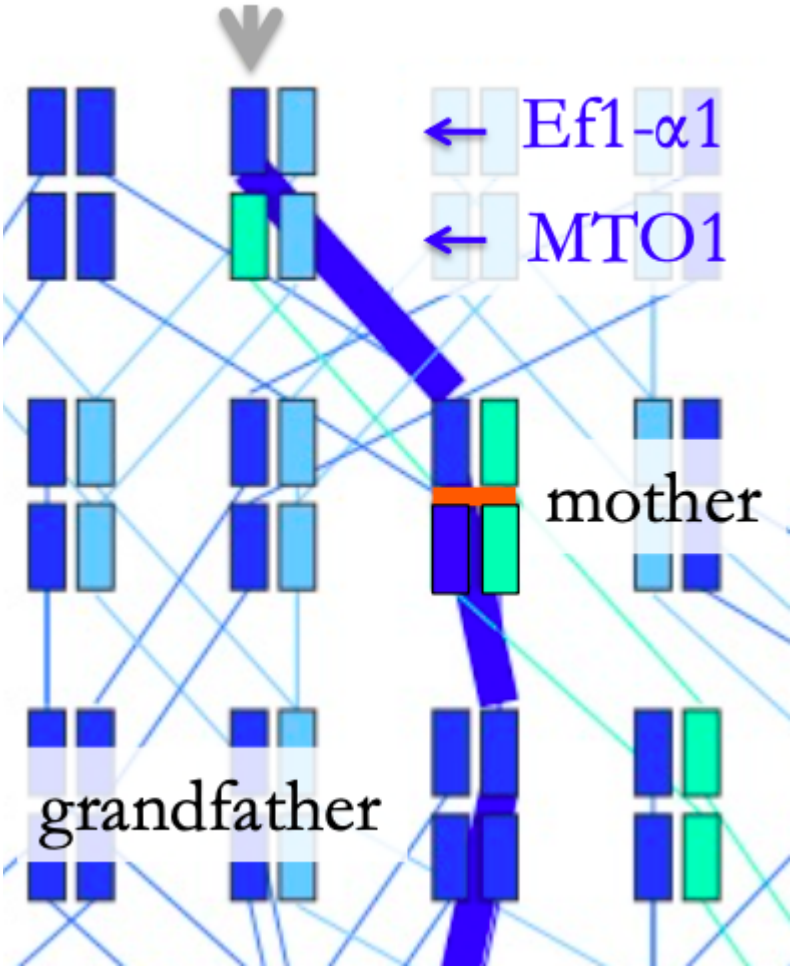


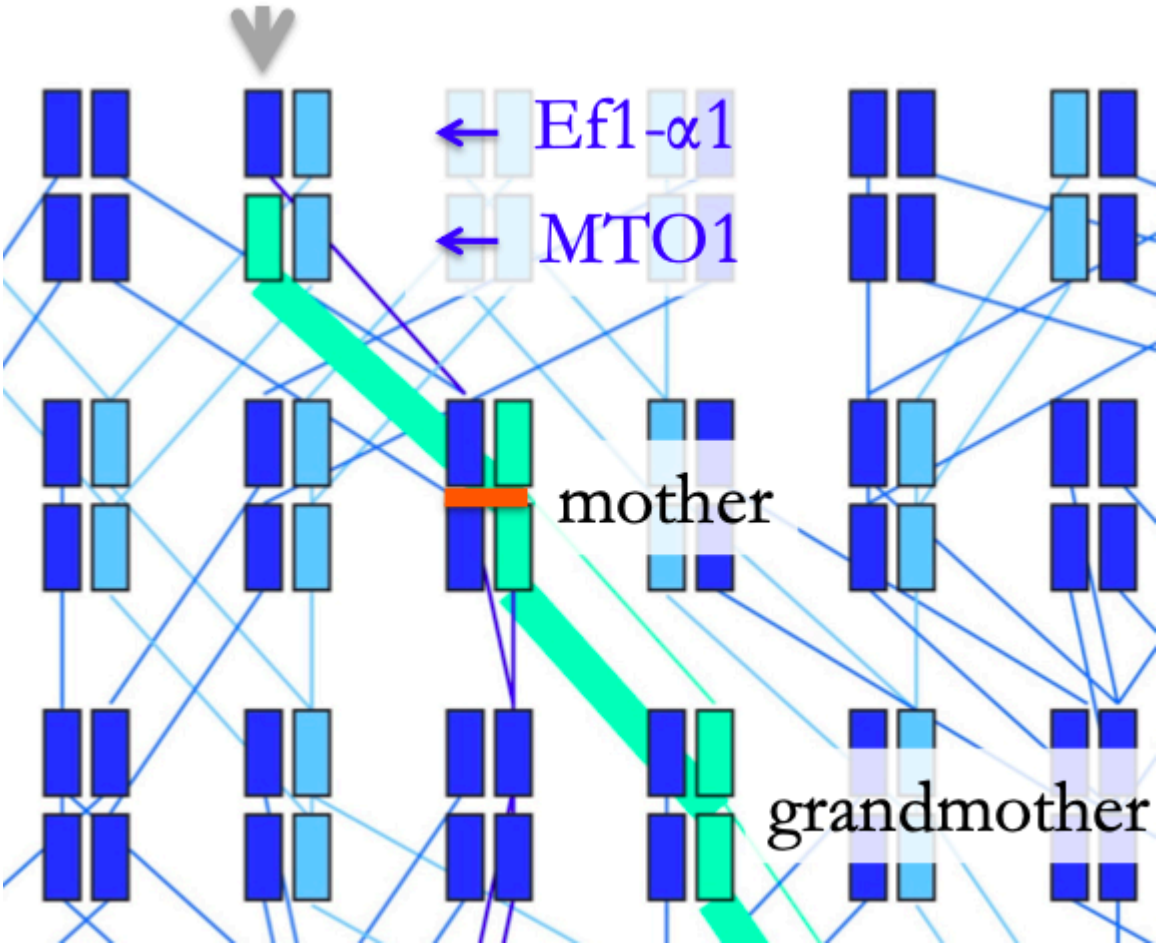
- The path of genetic descent of a genetic locus within a population has the form of a phylogenetic tree, a gene tree.
- It has the "form" of a tree, because of its branching history. Nodes represent a copy descending to multiple children. Looking backwards, two copies "coalesce" into one at these nodes.
- On average, the MRCA of all alleles is **~4N** generations ago (see Drift).
- Aside: The rate of coalescent events (coming together in nodes) provides a signal of past population sizes

②If recombination happens between the loci, the gamete has a mix of copies from both parents:


Ef1-α1 locus
MT01 locus
If recombination happens between the loci, the gamete has a mix of copies from both parents
make gamete
Chromosome from one parent
Chromosome from other parent


Ef1-α1
MTO1
recom


Ef1-α1
MTO1
mother
grandfather



Key points:

- Linkage binds adjacent sections of the genome to have the same gene tree
- Recombination breaks this up.
- Recombination can occur within genes, so different segments can have different trees.
- The further apart are two loci,the more different their gene trees are.
- There is no ‘single’ tree – we are collections of genes
- Coalescence looking backwards, drift and fixation looking forwards

③ Who are we (genetically)?

- Most of our genes differ little,due to recent common ancestry.
- While there are > 8 billion of us today,the rate of loss of genetic diversity in our evolutionary tree suggests a bottleneck of ~10,000 individuals

- Further genetic variation was lost due to repeated founder effects as humans spread across the globe
- This net result is that that humans are extremely closely related, descended from a very small population only ~200,000 years ago.
- On average, two individuals differ at only ~0.1% of their genomes.
- Most genetic variants found in one geographical region are found in all regions or at least in some other regions

The traits that make us human are inherited from our ancestors

- Many species of human walked the Earth at the same time
- EarlyHominidswereinAfrica,laterHomospecieswereglobal
- Genetic data tells us that Homo sapiens evolved in Africa, and then migrated globally ("Out of Africa") and that this was a population bottleneck event

G. Neanderthals

- Archaeological evidence suggests Homo sapiens and Neanderthals overlapped for at least 5,000 years

①Interbreed?

- 1820: Neanderthal fossils discovered in Neander valley, Germany
- 2003: Human genome project completed
- 2010: Neanderthal genome project completed in Leipzig, Germany

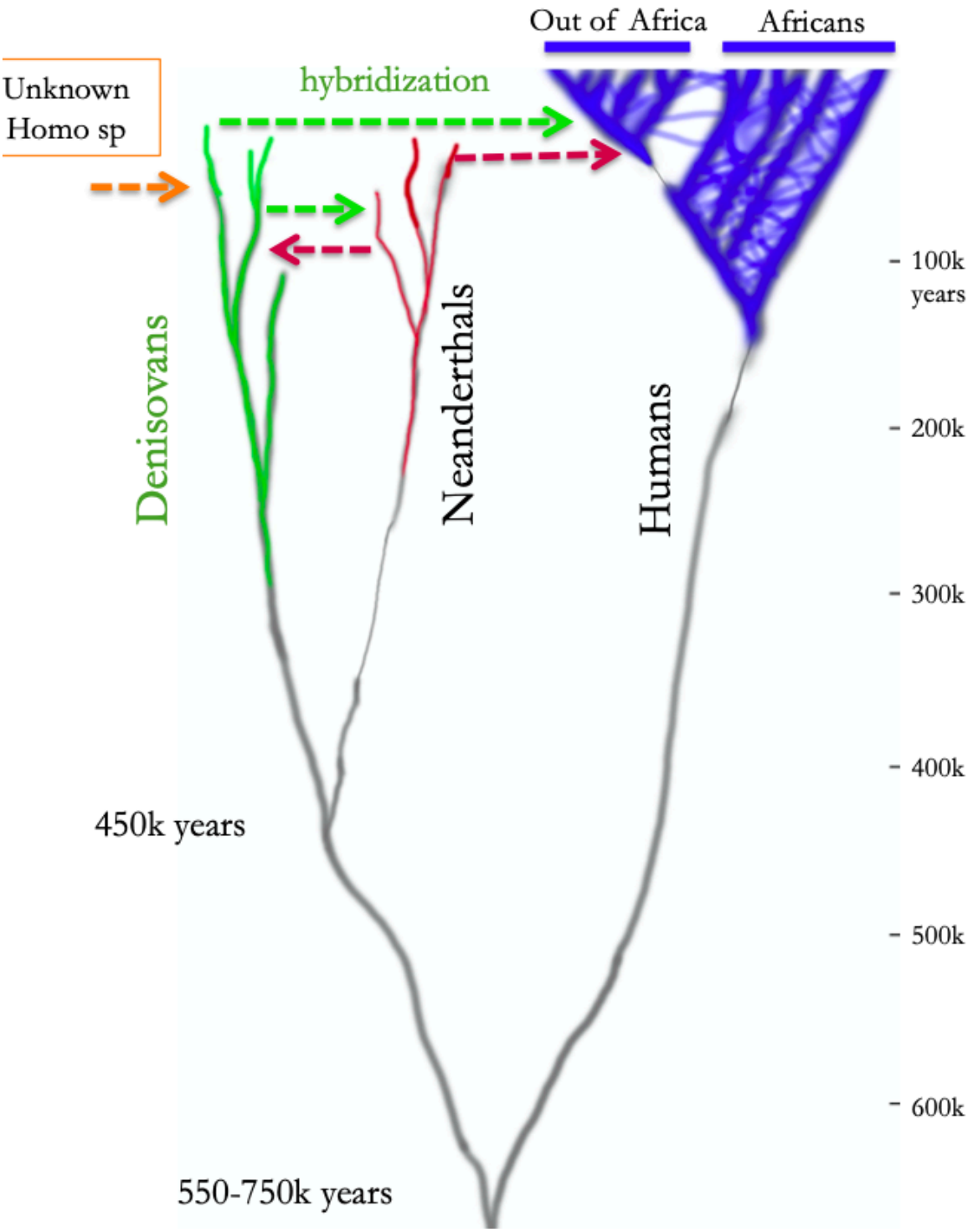
Result: Neanderthal ancestry account for 1-4%

- Hybridization with Neanderthals happened 200k in Eurasia, interbreeding with humans that had left Africa.
- For these regions, the MRCA goes back much further in time

② What do Neanderthal genes do?

- After introgression into humans, most Neanderthal genes were lost

    - Neanderthal ancestry lower on X
    - Genome-wide, Neanderthal ancestry decreasing in
    - Europeans over 40k years
    - Decrease stronger near genes
- FOXP2
    - Required for language
    - Human version has two amino acid changes, subject to a selective sweep ~200,000 years ago
    - Neanderthals share the same version
- Humans also hybridized with Denisovans
    - Traces of Denisovan genes are rarer (<0.1%) and mostly found among people in Oceania.



3. Evolution of Sex

A. eukaryotes:

① The vast majority of eukaryotes reproduce sexually, at least occasionally.

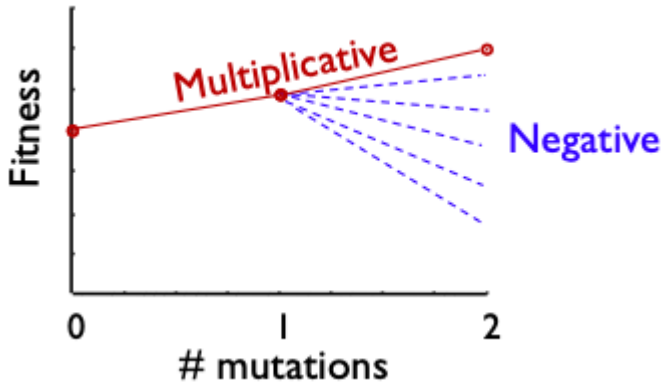
- Sexual Reproduction: Mating with another individual, leading to the production of genetically mixed offspring.
- Asexual Reproduction: Production of offspring without mating or genetic mixing, e.g., via tillers (strawberries), budding or fission (yeast), or asexual seeds or eggs (e.g., dandelions, whiptail lizards).
- All living eukaryotes, including exclusively asexual lineages , descend from a most recent common ancestor that was already sexual ~1-1.6BYA*

② reason of evolving sex

- Costs
    - Cost of producing males: Sexual females, by dividing resources between daughters and sons, should be quickly outcompeted by asexual variants (all else being equal).
    - Destruction of favorable gene combinations
    - Cost of finding a mate
    - Risk of not finding a mate
    - Risk of disease transmission during mating
- Advantages
    - sex can generate variation needed by selection through recombination and segregation.
    - But:
        - 1) Sex need not increase variation
        - 2) Not all variation helps selection
            - Segregation load (within loci): AA → Aa
            - Recombination load (between loci): AB → Ab/aB

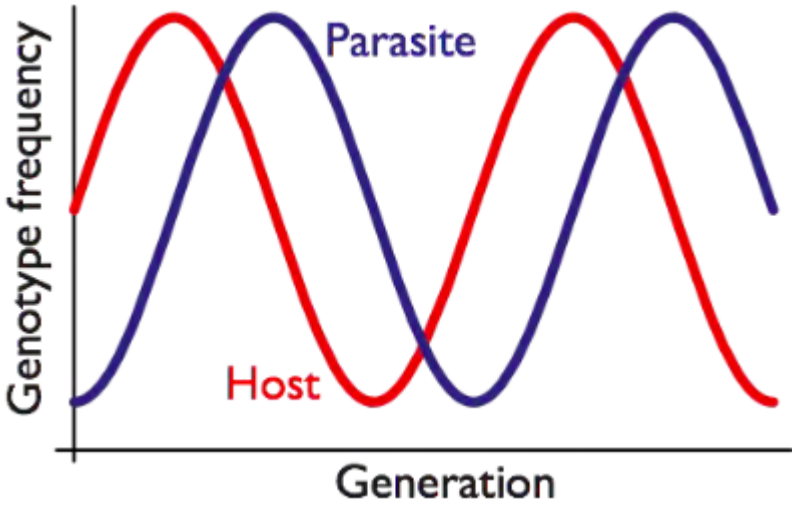
B. Negative epistasis hypothesis for sex

- If fitness interactions among genes were predominantly negative



  - Extreme genotypes would be less fit, on average
  - Extreme genotypes would become less frequent, decreasing genetic variance in fitness through disequilibrium
  - Sex could restore genetic variance & improve response to selection

C. Red Queen hypothesis for sex



- sexual reproduction has evolved to increase the rate of adaptation in the face of changing interactions with other species (e.g., host-parasite interactions).

- As long as there is ample genetic mixing within a population and selection is not too strong, modifier models show that the frequency of sex often decreases as a result of the Red Queen.
- Sex mainly breaks apart the fit gene combinations that allowed the parents to survive species interactions

D. clonal interference: Fisher-Muller Hypothesis

- Beneficial mutations often appear in different individuals.
- In asexual reproduction, these mutations compete with each other, resulting in low efficiency.
- Sexual reproduction allows recombination to combine beneficial mutations, accelerating adaptation.

Muller's ratchet:

- In asexual populations, drift randomly eliminates the least mutated genotype, and the lack of recombination prevents its regeneration.
- The results are:
    - Irreversible accumulation of harmful mutations
    - Decrease in average fitness
- Sexual reproduction and recombination prevent ratcheting

E. sex ratios

- Sex ratios among sexually reproducing species are very often near 1:1

Fisher's Sex Ratio Theory

- Offspring inherit half their genes from the father and half from the mother → Reproductive value must be equal for both sexes.
- If a particular sex is scarce, that sex has a higher reproductive success rate.
- Parents tend to produce the scarce sex to increase their own gene transmission.

- Natural selection drives the sex ratio back to 1:1.
- The sex ratio is determined by the balance of parental expenditure.

F. Summary

- There is incredible diversity in how organisms reproduce.
- Nevertheless, the majority of eukaryotes reproduce sexually at least every once in a while.
- All living eukaryotes descend from an ancestor that already had the genetic tools for sex.
- Sex is, however, costly – including the costs of dividing resources between males and females and the costs of breaking apart gene combinations that work in the current environment.
- The most promising set of models hypothesise that sex has evolved to untie alleles from the genomes in which they are found, thereby reducing selective interference among loci.
- While female-biased sex ratios would reduce the cost of sex, the rarer sex typically contributes more to the next generation, selecting for mutations that equalize the sex ratio.